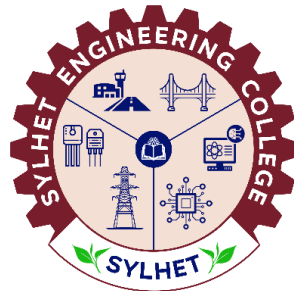


A Thesis Submitted to the Sylhet Engineering College for the Degree of  
**Bachelor of Science in Electrical and Electronic Engineering**

# **Hyperbolic Metamaterial Induced Optical Pulling of Single and Clustered Nanorods with Distinct Properties**

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July 2025  
Sylhet Engineering College, Sylhet  
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The thesis titled **“Hyperbolic Metamaterial Induced Optical Pulling of Single and Clustered Nanorods with Distinct Properties”** was submitted by **Md. Al Amin Chy**; Student ID: **2019338509** respectively; Session **2019-2020**, to the Department of Electrical and Electronic Engineering, Sylhet Engineering College, Shahjalal University of Science and Technology, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Electrical and Electronic Engineering and approved as to its style and contents.

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# Acknowledgements

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We begin by extending our gratitude to the Almighty, whose guidance and sustenance have enabled us to embark on this journey and complete the requirements for the Bachelor of Science in Electrical and Electronics Engineering (EEE), including our thesis work. It is through His grace that we have attained the strength and determination to achieve this milestone.

We want to express our sincere appreciation to our esteemed supervisor, **Md. Janibul Alam Soeb**, Assistant Professor, Department of Firm Power and Machinery, Sylhet Agricultural University, Sylhet. His invaluable academic mentorship, exemplary leadership, positive encouragement, insightful advice, and unwavering support have been instrumental throughout our academic pursuit. We are truly grateful for his guidance. Furthermore, we extend our thanks to our other faculty members, **Md. Shahid Iqbal**, **Salman Fazle Rabby**, **Md. Ashraful Alam** and **Mahedi Kamal Ahmed** for their kindness, support, and cooperation during our academic journey. Their assistance has been greatly appreciated.

Our heartfelt gratitude also goes to our parents, siblings, classmates, and friends in the EEE department at Sylhet Engineering College. We acknowledge the sacrifices, prayers, encouragement, and unwavering support they have provided, which have contributed significantly to our success. We are indebted to them for their role in shaping our academic and personal development.

# Abstract

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In the field of optical manipulation, the generation of optical pulling forces using hyperbolic metamaterials (HMMs) has emerged as a promising technique for counterintuitive light-matter interactions. This study investigates the optical pulling behavior of plasmonic, chiral, and dielectric nanorods positioned on an HMM under a time-harmonic plane-polarized laser beam incident along the “+Z” axis to the “-Z” axis direction. This research is divided into two primary configurations. In the first case, plasmonic, chiral, and dielectric nanorods are arranged in a heterogeneous cluster over the HMMs and analyzed for the optical force response under three environmental conditions: fully immersed in water, partially immersed (half air-half water), and fully in air. Full-wave simulations reveal that in all three environments, all the objects (plasmonic, chiral, dielectric) nanorods consistently experience an optical pulling force at the  $10^{-20}$  N scales for clustered configuration. In the second case, each nanorod type was independently placed on the same HMMs substrate within a full-air medium and investigated their force behaviors, while the plasmonic experience 100 times more than others for the single setup, which is  $10^{-19}$  N ranges. Like the case of the cluster, three nanorods alone demonstrate a stable optical pulling force, also getting the pulling force with the variation of length, radius, and distance between them, which makes this optical setup more practical. It is important to note that plasmonic behavior is highly sensitive to the spatial arrangement of the nanostructures, whereas the optical response of dielectric materials is more dependent on their physical length. This study highlights the robustness of optical pulling force generation, as aforementioned, all nanorods using a simplified HMMs structure, and offers a pathway for scalable optical manipulation of anisotropic nanostructures, particularly in sensing and nanophotonic applications.

**Keywords:** *optical manipulation, pulling force, Hyperbolic Metamaterials, chirality, plasmonic.*

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# List of Abbreviations

| Abbreviations       | Full Form                   |
|---------------------|-----------------------------|
| Au                  | Gold                        |
| B                   | Magnetic Flux Density       |
| c                   | Speed of Light              |
| COMSOL              | COMSOL Multiphysics         |
| D                   | Electric Displacement Field |
| EMT                 | Effective Medium Theory     |
| E                   | Electric Field              |
| f                   | Filling Ratio               |
| H                   | Magnetic Field              |
| HMM                 | Hyperbolic Metamaterial     |
| $\bar{I}$           | Identity Tensor             |
| k                   | Wavenumber / Wavevector     |
| L                   | Length (of Nanorod)         |
| Li                  | Depolarization Factor       |
| MST                 | Minkowski Stress Tensor     |
| m                   | Magnetic Dipole Moment      |
| $\mu$               | Permeability                |
| NR                  | Nanorod                     |
| n                   | Refractive Index            |
| OPF                 | Optical Pulling Force       |
| OBF                 | Optical Binding Force       |
| OTB                 | Optical Tractor Beam        |
| PDOS                | Photonic Density of States  |
| PML                 | Perfectly Matched Layer     |
| P                   | Plasmonic Layer             |
| R                   | Radius (of Nanorod)         |
| RF                  | Radio Frequency             |
| SiO <sub>2</sub>    | Silicon Dioxide             |
| SPP                 | Surface Plasmon Polariton   |
| SPR                 | Surface Plasmon Resonance   |
| $\bar{T}$           | Stress Tensor               |
| TM                  | Transverse Magnetic         |
| $\nabla$            | Gradient Operator           |
| $\langle F \rangle$ | Time-Averaged Optical Force |
| $\alpha_e$          | Electric Polarizability     |
| $\alpha_m$          | Magnetic Polarizability     |



# Chapter 1: Introduction

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This research investigates the generation of optical pulling forces on nanorods using a hyperbolic metamaterial (HMM) composed of alternating plasmonic and dielectric layers. Two main configurations were explored: a heterogeneous cluster of plasmonic, chiral, and dielectric nanorods, and individual nanorods placed separately on the HMMs. The system was tested under various surrounding media such as air, half air-half water, and fully submerged in water with a plane-polarized laser incident along the  $+Z$  to  $-Z$  direction. In all cases, all the nanorods consistently exhibited a strong optical pulling force. Full-wave simulations confirmed the effectiveness of the HMMs in enabling near-field optical pulling. This study highlights the simplicity and robustness of the setup for targeted manipulation of nanostructures. The findings have potential applications in nanorobotics, pharmaceutical industries, and photonic device development.

## 1.1 Optical Manipulation

Optical Manipulation refers to the use of light to control and move microscopic and nanoscopic structures without physical contact. This technique relies on the momentum transfer from photons to matter, enabling forces that can trap, push, or pull small objects. First demonstrated by Arthur Ashkin in the form of optical tweezers, optical manipulation has revolutionized fields like biology, nanotechnology, and materials science. It allows precise handling of particles such as cells, nanoparticles, and molecules. Optical forces are typically categorized into scattering, gradient, and, in some cases, pulling forces. Advanced materials like metamaterials and photonic crystals have expanded the possibilities of this technique. Today, optical manipulation plays a vital role in non-invasive sensing, particle sorting, and nanoscale assembly.

## 1.2 Significance of optical pulling

Optical pulling offers a groundbreaking approach to non-contact manipulation at the nanoscale, enabling precise control over particles using light. This phenomenon is particularly useful in biomedical applications, such as targeted drug delivery and single-cell manipulation. In environmental engineering, it can assist in the extraction of microplastics or contaminants from

water without chemical agents. Optical pulling also plays a vital role in nanofabrication and the assembly of nanodevices with high spatial accuracy. Its ability to control diverse nanostructure types, including dielectric, plasmonic, and chiral, makes it highly versatile for both research and industrial applications.

### **1.3 Metamaterial & Hyperbolic Metamaterial**

Metamaterials are artificially structured materials engineered to control electromagnetic waves in ways that are not possible with natural materials, enabling phenomena like negative refraction, cloaking, and subwavelength focusing. They are typically made from periodic arrangements of subwavelength units like metal-dielectric composites. Their properties depend more on structure than composition, offering tunability for specific optical, acoustic, or thermal functions.

Hyperbolic Metamaterials (HMMs) are a type of anisotropic metamaterial with opposite signs of permittivity in orthogonal directions, leading to hyperbolic dispersion. This enables them to support highly confined, high-momentum electromagnetic waves, which are not allowed in conventional materials. HMMs are widely used in nano-imaging, spontaneous emission control, and optical pulling force generation due to their unique light guiding abilities.

### **1.4 Role of Hyperbolic Metamaterial**

Hyperbolic metamaterials (HMMs) play a transformative role in the field of optical manipulation by enabling highly controlled light–matter interactions at the nanoscale. Unlike conventional materials, HMMs exhibit extreme anisotropy, characterized by having opposite signs of permittivity along orthogonal axes. This results in a unique hyperbolic dispersion relation, which allows them to support unbounded high- $k$  propagating modes and highly confined electromagnetic waves. These properties enhance the local photonic density of states, enabling stronger optical forces.

In optical manipulation, HMMs can be used to generate optical pulling forces, where nanostructure objects are drawn toward the light source contrary to the usual pushing effect of radiation pressure. This phenomenon is especially useful for manipulating nanostructure objects, including dielectric, plasmonic, and chiral nanorods. By placing such objects above a hyperbolic substrate and illuminating them with a plane-polarized light beam, HMMs facilitate

the redirection of light momentum and induce pulling through enhanced scattering and interference effects.

Moreover, HMMs amplify near-field interactions and enable stable optical trapping and sorting of heterogeneous objects without requiring complex beam shaping. They also support the formation of multipolar modes in dielectric nanorods and enable phenomena like Fano resonance and electric–magnetic dipole interference in plasmonic and chiral nanorods. This versatility makes HMMs highly suitable for applications in biomedical manipulation, nanoparticle assembly, and fluidic transport systems. Overall, HMMs serve as a powerful platform for achieving precise, non-contact control in advanced optical manipulation systems.

## 1.5 Objectives

- To investigate and validate the optical pulling effect (tractor beam phenomenon) on plasmonic, chiral, and dielectric nanorods placed on a Hyperbolic Metamaterial (HMM) substrate under varying environmental conditions through full-wave electromagnetic simulations.
- To understand the underlying physical mechanisms responsible for optical pulling by employing Minkowski’s Stress Tensor (MST), dipole interaction theory, and detailed analysis of current density and electromagnetic field distributions.

## 1.6 Thesis Structure

**Chapter 1:** This chapter introduced the concept of optical manipulation and highlighted the significance of optical pulling at the nanoscale. It discussed the unique role of hyperbolic metamaterials (HMMs) in enhancing light–matter interactions and enabling strong pulling forces. Finally, it outlined the research objectives focused on exploring and understanding optical pulling in various nanorod configurations using HMM substrates.

**Chapter 2:** This chapter reviews a wide range of optical manipulation techniques, emphasizing structured light, metamaterials, and HMM-based setups for generating optical pulling forces on various nanoparticles. It highlights how the proposed work differs by achieving consistent pulling across multiple particle types and media using a simplified, robust nanorod-HMM configuration.

**Chapter 3:** This unit explains the theoretical principles behind optical forces, including gradient, scattering, and absorption forces, with a focus on Minkowski's Stress Tensor for accurate force calculations. It further analyzes the properties of nanorods, HMM behavior, and the optical force mechanisms acting on both single and clustered nanorods under electromagnetic excitation.

**Chapter 4:** This chapter presents a comprehensive simulation-based methodology using COMSOL Multiphysics to study optical pulling on plasmonic, chiral, and dielectric nanorods over a multilayer HMM substrate. It outlines substrate design, material parameters, nanorod configurations, environmental setups, and the use of Minkowski's Stress Tensor for accurate optical force computation.

**Chapter 5:** This chapter presents detailed simulation results confirming stable optical pulling forces on both single and clustered nanorods across various media, configurations, and structural parameters. It further analyzes the physical mechanisms such as field distributions, dipole interactions, and current density reversal that enables optical pulling in chiral, plasmonic, and dielectric nanorods.

**Chapter 6:** Concluding the study, this chapter highlights the successful demonstration of optical pulling forces on diverse nanorods using a simplified HMM-based setup under different conditions. It also outlines potential extensions of the work, including dynamic particle control, tunable substrates, and integration into advanced optical manipulation systems.

## **1.7 Summary**

This chapter introduces the concept of optical pulling forces on nanorods using hyperbolic metamaterials (HMMs), highlighting their unique ability to enable precise, non-contact manipulation across various nanorod types and environments. It outlines the significance, theoretical background, objectives, and thesis structure for exploring optical pulling through full-wave simulations and advanced analysis methods.

## Chapter 2: Literature Review

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There are different methods that have been developed for optical manipulation and generating optical pulling forces, including optical tweezers, structured light, metamaterials, and hybrid systems. Optical tweezers use gradient forces from focused beams to trap particles. Structured light beams like Bessel or vortex beams can shape momentum flow for advanced control. Hyperbolic metamaterials enhance near-field interactions and support pulling by redirecting optical momentum. Hybrid systems combine material design and beam shaping for versatile, non-contact manipulation at the nanoscale.

### 2.1 Related Works

**Mostafizur Rahman Rana** et al.[1] developed a generic optical manipulation method using a hyperbolic metamaterial (HMM) platform to generate optical pulling forces on nanoparticles. The technique uses a time-harmonic plane-polarized laser interacting with a multilayered chiral-plasmonic HMM structure. It can pull Rayleigh-sized dielectric, plasmonic, and chiral nanoparticles individually or in clusters. Simulations in air and water environments show the setup's robustness. Validation was done by analyzing current density, field distribution, and forces from Fano resonance and dipole interference.

**Renchao Jin** et al. [2] introduced a method to generate optical pulling forces on gold nanowires using hyperbolic metamaterials (HMMs). The technique exploits the concave isofrequency contour of HMMs and directional surface plasmon polaritons (SPPs) to achieve broadband pulling forces under plane wave illumination. Simulations show the method works for nanowires embedded in or above gold-water or gold-silica multilayer HMMs, with wide wavelength and angle insensitivity. The approach's effectiveness is confirmed by comparing forces to those on a gold substrate. The study highlights dual pulling mechanisms from topology-momentum and SPP effects for nanoscale manipulation.

**Ufuk Parali** et al. [3] proposed a method for optical levitation and trapping of transparent microspheres using Type-I hyperbolic metamaterials (HMMs). The structure, made of alternating Ge-AZO layers, enhances optical forces under tightly focused laser beams. Using ray-optics approximation, the study shows significantly stronger axial forces and better radial

stability than conventional dielectrics. “Zero-force” paths are demonstrated for TEM<sub>00</sub> and TEM<sub>01</sub>\* beams. The method also shows robustness against thermal fluctuations modeled via the Langevin equation.

**Andrey A. Bogdanov** et al. [4] introduced a method for enhanced optomechanical manipulation using a 3D nanorod metamaterial with hyperbolic dispersion. The technique uses near-field interactions to trap and manipulate gold nanoparticles under laser illumination. Optical forces are calculated via full-wave simulations and a dipolar model. Results show strong in-plane trapping and pulling forces ( $\sim 2.3 \times 10^3$  pN/W/nm<sup>3</sup>). The method performs robustly across elliptic, hyperbolic, and ENZ regimes, with minimal impact from topological transitions.

**Alexander S. Shalin** et al. [5] proposed a method for optical manipulation using hyperbolic metamaterials to generate pulling forces on nanoparticles. The technique excites confined extraordinary modes via scattering from an interface imperfection, creating optical attraction for gold and dielectric particles. The method is applied to a metamaterial slab to achieve sub diffraction trapping. Its effectiveness is validated by numerical analysis showing robust pulling forces across different polarizations and particle materials.

**Mostafizur Rahman Rana** et al. [6] proposed a generic optical setup for an optical tractor beam to pull clusters of plasmonic, dielectric, and chiral Mie objects. The method uses two time-harmonic plane-polarized laser beams over a dielectric substrate, achieving pulling forces via multipolar interactions and Lorentz force reversal. The setup is effective for wavelengths between 660–860 nm, validated through detailed momentum contribution analysis. This simple, chemical-free approach has potential for industrial nanoparticle extraction. The technique demonstrates robust pulling across diverse particle types and sizes.

**Saikat Chandra Das** et al. [7] proposed a near-field optical sorting method for plasmonic, dielectric, and chiral nanoparticles using a linearly polarized plane wave at a 45° angle. The technique achieves sorting by inducing distinct optical forces: pulling for dielectric, pushing for plasmonic, and lateral for chiral particles. Three setups (air, half-immersed, and fully immersed in water) support sorting Rayleigh-sized particles. The method leverages plasmonic resonances and magnetic dipoles, offering a chemical-free approach for industrial nanoparticle separation. Its effectiveness is validated across 600–1200 nm wavelengths.

**T. V. Raziman and Olivier J. F. Martin** et.al [8] proposed a computational study of internal optical forces in plasmonic nanostructures, including gap antennas, dolmen structures, and split rings. The method reveals strong lateral forces perpendicular to light propagation, driven by polarization charges. These forces exhibit rich behavior with varying wavelength, polarization, and geometry, enabling applications in optical manipulation. The approach uses surface integral equation simulations and Minkowski's stress tensor for force calculations. Its effectiveness is demonstrated across different plasmonic systems, highlighting force reversals and negative torques.

**Jannatul Shahrin Shoshi** et al. [9] proposed an on-chip all-optical method to distinguish single Rayleigh particles (plasmonic, dielectric, chiral) using distinct optical forces. A linearly polarized plane wave on a plasmonic substrate induces pushing, pulling, and lateral forces, respectively, enabling sorting without complex beams or chemicals. The technique, validated via COMSOL simulations, works for independently placed particles but not clusters. It offers a sustainable alternative for industrial nanoparticle sorting.

**Sudipta Biswas** et al. [10] proposed a method to control the counterintuitive repulsive thrust of plasmonic-chiral nano-dimers using counter-propagating plane waves in homogeneous and heterogeneous media. The technique leverages modified longitudinal Optical Binding Force (OBF) to achieve repulsion in near and far-field regions. In a homogeneous (air) background, the wave nature of light dominates, while at an air-water interface, Minkowski momentum transfer supports particle-like behavior. The method, validated via COMSOL simulations, offers a novel approach for nanoparticle manipulation.

**E. Lee and T. Luo** et.al [11] proposed a method for long-distance optical pulling of a nanoparticle (NP) in a low-index cavity using a single plane wave. The NP-in-cavity structure, with a cavity refractive index lower than the medium, enables optical pulling force (OPF) via unique internal electromagnetic modes. Using a silica-Au core-shell NP encapsulated by a plasmonic nanobubble, they experimentally demonstrated pulling an Au NP over ~0.1 mm. The approach, validated through simulations and experiments, offers potential for nanostructure assembly and nano-swimmers.

**S. Sukhov and A. Dogariu** et.al [12] proposed a novel concept of optical "tractor beams" using nondiffracting, rotating scale-invariant vector electromagnetic waves. Their method demonstrates that nonconservative forces can be oriented opposite to the beam propagation,

enabling objects to move toward the light source. The approach is validated through theoretical analysis and simulations, showing negative optical force regions for small particles. The effectiveness is illustrated with a silver nanoparticle in carbon tetrachloride, achieving stable trapping with 100mW illumination power. This work advances optical manipulation techniques beyond traditional optical traps.

**Jun Chen** et al. [13] introduced a method to achieve an optical pulling force (OPF) using Bessel beams, enabling particles to be pulled toward the light source. The technique relies on maximizing forward scattering through multipole interference in dielectric particles. Validated numerically with polystyrene beads, OPF occurs at specific particle sizes with stable transverse trapping. The approach leverages large cone angles and is effective in the Mie regime. This advances optical micromanipulation by adding pulling functionality.

**Huajin Chen** et al. [14] demonstrated that Fano resonance induces a negative optical scattering force (NOSF) on plasmonic nanoparticles using a zeroth-order Bessel beam. The NOSF arises from enhanced forward scattering due to multipole interference, observed in core-shell, homogeneous, and hollow metallic particles. Extinction spectra confirm Fano resonance from dipole-quadrupole or quadrupole-octupole interactions. This tunable NOSF enhances optical micromanipulation, enabling size-selective trapping and sorting of plasmonic nanoparticles.

**Alexander Poddubny** et al. [15] reviewed hyperbolic metamaterials, artificial media with subwavelength structures exhibiting hyperbolic dispersion due to opposing signs in their permittivity tensor components. These materials enable unique properties like enhanced spontaneous emission, diverging density of states, negative refraction, and superlensing. Common realizations include layered metal-dielectric structures and nanowire arrays. Experimental demonstrations show significant Purcell factor enhancements near the elliptic-to-hyperbolic transition. Applications include advanced optical micromanipulation and subdiffraction imaging.

**K. Saffat** et.al [16] proposed an all-optical sorting technique for nanoscale Mie particles using a Metasurface with double-lined gold nanostructures. The setup generates dual surface plasmon polariton (SPP)-induced plasmonic fields, enabling distinct behaviors: plasmonic particles experience optical pulling force, dielectric particles experience pushing force, and chiral particles experience lateral force. This method distinguishes silica, gold, and chiral nanospheres without chemicals. The proposed on-chip approach offers a cost-effective solution



for material-based optical sorting. Potential applications include advanced nanoscale manipulation and sorting.

**Dongliang Gao** et al. [17] proposed a method to achieve tunable optical pulling and pushing forces on active plasmonic nanoparticles at Fano resonance. The nanostructure, with a gain core and metallic shell, exhibits enhanced forces due to strong field amplification. Fano resonance arises from coupling between quadrupole and dipole modes, enabling force direction reversal. The Lorentz force on bound currents dominates the giant negative forces. This approach offers insights into light-matter interactions and nanoparticle manipulation.

**Tongtong Zhu** et al. [18] proposed an optical manipulation method using attenuated modes in subwavelength photonic crystal waveguides. The technique enables both pulling and pushing forces along the channel, overcoming Brownian motion within a critical length. By tuning the damping rate, the force direction can be controlled for sorting objects by size or refractive index. The method supports curved trajectory manipulation without complex beams. It offers a flexible approach for optical sorting and extraction in various channel types.

**MM Rahman** et.al [19] proposed an active tractor beam mechanism using two superimposed higher-order Bessel beams with reverse helical nature and different frequencies. The setup enables stable long-distance levitation, controlled rotation, and 3D trapping of multiple immersed objects. Periodic intensity variations drive bidirectional particle movement without manual phase ramping. The method allows manipulation of objects of varying sizes by adjusting binding region dimensions. It offers a versatile single optical setup for advanced particle control.

**Onofrio M. Maragò** et al. [20] reviewed optical trapping and manipulation techniques for nanostructures, focusing on the nanoscale range. They discussed methods like plasmon-enhanced forces and holographic optical tweezers for manipulating nanoparticles and nanowires. The study highlights the use of plasmonic resonances to enhance optical forces, enabling stable trapping at lower powers. Techniques such as T-matrix and discrete dipole approximation are employed to calculate forces beyond the dipole approximation. The review emphasizes applications in assembling complex nanostructures for nanomaterials-based technologies.

**Aristide Dogariu** et al. [21] explored optically induced 'negative forces' in a 2013 Nature Photonics article. They reviewed methods to achieve negative mobility using optical beams, including tailoring environments with metamaterials or leveraging interactions between

particles. Techniques like overlapping Bessel beams and exploiting plasmonic resonances enable particle movement opposite to light propagation. The study highlights applications in biology, chemistry, and nanotechnology, emphasizing non-conservative force manipulation.

**Hang Li** et al. [22] reviewed optical pulling forces, a novel counterintuitive optical manipulation technique. They explored mechanisms like structured environments and particle interactions to induce negative mobility. The study includes related phenomena such as left-handed optical torque and lateral optical forces. Potential applications span remote mass transport, optical rotation, and sorting. Analogous effects in acoustic, water, and quantum waves are also examined.

**Alexander S. Shalin** et al. [23] proposed a plasmonic nano-accelerator using V-groove nanostructures to propel nanoparticles. The model leverages field enhancement in metallic grooves to achieve velocities up to tens of centimeters per second. It employs low-intensity optical fields ( $1\text{--}2\text{ mW}/\mu\text{m}^2$ ) for directional acceleration. Composite core-shell nanoparticles with negative polarizability enable ejection in the visible spectrum. Potential applications include medical delivery and nanolithography.

**A. V. Kabashin** et al. [24] introduced a plasmonic nanorod metamaterial for enhanced biosensing. The porous nanorod layer supports a guided mode with high sensitivity to refractive-index changes ( $>30,000\text{ nm/RIU}$ ). Using a streptavidin-biotin model, it detects small analytes (e.g.,  $244\text{ Da}$  biotin) at sub-micromolar levels. The metamaterial's nanoporous structure enables size selectivity and increased biomaterial incorporation. Its compatibility with SPR equipment supports applications in biological and chemical sensing.

**Anthony J. Hoffman** et al. [25] demonstrated negative refraction in semiconductor metamaterials using InGaAs/InAlAs heterostructures. The material exhibits a negative refractive index in the mid-infrared ( $8.8\text{--}13.1\text{ }\mu\text{m}$ ) with a figure of merit up to 330. Experimental reflection and transmission measurements confirm the transition from positive to negative refraction.

## 2.2 Summary

This chapter reviews various optical manipulation methods, emphasizing hyperbolic metamaterials and advanced beam techniques for generating optical pulling forces on diverse nanostructures. It highlights recent research demonstrating robust, versatile pulling across

different particle types and media. The proposed nanorod-HMM setup uniquely achieves consistent near-field enhanced pulling in multiple environments, suitable for practical applications.

## Chapter 3: Theoretical Analysis

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Optical forces are generated when light transfers momentum to matter, allowing the manipulation of microscopic and nanoscopic particles. These forces, classified as gradient, scattering, and absorption, enable key applications such as optical trapping, levitation, and pulling. To calculate these forces accurately, especially in complex environments, Minkowski's Stress Tensor is used, providing a reliable method for evaluating the total electromagnetic force on nanostructures like nanorods placed near hyperbolic metamaterials.

### 3.1 Principle of Optical Forces

Optical forces arise due to the exchange of momentum between light and matter, a fundamental concept governed by Minkowski's equations and the conservation of momentum. When electromagnetic waves interact with matter such as nanoparticles, atoms, or molecules, the light can exert forces that either push, pull, or trap these objects. These forces are categorized based on the nature of interaction: gradient force, scattering force, and absorption/radiation pressure force.

These forces are foundational in technologies like optical tweezers, optical levitation, nano-manipulation, and, more recently, in optical pulling or tractor beams.

#### Types of Optical Forces:

##### (a) Gradient Force

The gradient force arises from the spatial variation of the electromagnetic field intensity. It pulls a particle toward the region of highest intensity and is responsible for stable trapping (as in optical tweezers).

$$F_{\text{grad}} = \frac{1}{2} \text{Re}\{\alpha \nabla |E|^2\} \quad (1)$$

where  $\alpha$  is complex polarizability,  $E$  local electric field,  $\nabla |E|^2$  represents intensity gradient.

##### (b) Scattering Force

The scattering force (or radiation pressure force) results from the momentum transfer from photons to the particle. It acts in the direction of the incident beam and is associated with the re-radiated or scattered light.

$$F_{\text{scat}} = \frac{np_{\text{scat}}}{c} \hat{k} \quad (2)$$

where  $n$  is the refractive index,  $P_{\text{scat}}$  scattered power,  $C$  speed of light,  $K$  propagation direction.

### (c) Absorption Force

If the particle absorbs light, the absorption force also contributes to the net optical force. It is generally added to the scattering force as:

$$F_{\text{abs}} = \frac{np_{\text{abs}}}{c} \hat{k} \quad (3)$$

where  $P_{\text{abs}}$  is the absorption power. The total time-averaged optical force acting on a small particle in a light field is:

$$\langle F \rangle = \langle F_{\text{grad}} \rangle + \langle F_{\text{scat}} \rangle + \langle F_{\text{abs}} \rangle \quad (4)$$

In systems with multiple interactions (clusters, near-field substrates like HMM), the Minkowski Stress Tensor (MST) is used to calculate the optical force.

## 3.2 Minkowski's Stress Tensor

The Minkowski Stress Tensor (MST) is a fundamental tensorial formulation used to compute the electromagnetic force and torque exerted by fields on objects. In optical manipulation, especially when dealing with nanoparticles, photonic structures, and metamaterials, MST provides a rigorous and comprehensive method for evaluating time-averaged optical forces. Unlike simple dipole approximations, MST accounts for all components of the electromagnetic field incident, scattered, and reflected and is valid for particles of arbitrary size, shape, and material.

Electromagnetic waves carry both energy and momentum. When light interacts with a material object, part of this momentum is transferred to the object, resulting in a mechanical force. The net force on an object embedded in an electromagnetic field can be calculated by integrating the electromagnetic momentum flux (represented by MST) over a closed surface surrounding the object. The time-averaged force  $F$  exerted on a particle is obtained by integrating the Minkowski stress tensor  $T$  over a closed surface  $S$  enclosing the object:

$$F = \oint_S T \cdot \hat{n} dS \quad (5)$$

where  $F$  is the total force is Minkowski's stress tensor,  $n$  is a unit normal vector to  $S$ , and  $dS$  is a differential surface element.

### 3.3 Properties of Nanorods

Nanorods are elongated nanoscale structures characterized by a length much greater than their width, typically with an aspect ratio (length/diameter) ranging from 2:1 to 10:1. These structures exhibit unique physical, chemical, and optical properties that are distinctly different from their bulk or spherical counterparts due to their anisotropic geometry and quantum confinement effects. Nanorods can be synthesized from various materials, including metals (like gold and silver), semiconductors (such as CdSe, ZnO), and dielectrics, depending on the desired application. A key feature of nanorods is their shape anisotropy, which leads to direction-dependent properties. In metallic nanorods, the collective oscillation of free electrons in response to light, called surface plasmon resonance (SPR), is highly sensitive to the aspect ratio. Two distinct plasmon modes emerge: longitudinal (along the length) and transverse (across the width).

The resonance condition is given by:

$$\text{Re}[\varepsilon(\omega)] = -\frac{1-L_i}{L_i} \varepsilon_m \quad (6)$$

where  $\varepsilon(\omega)$  is the material permittivity,  $\varepsilon_m$  is the surrounding medium permittivity, and  $L_i$  is the depolarization factor.

In semiconductor nanorods, quantum confinement effects become significant when the dimensions approach the exciton Bohr radius, causing a size-dependent shift in the bandgap:

$$E_{\text{gap}}(L) = E_{\text{bulk}} + \frac{\hbar^2 \pi^2}{2\mu L^2} \quad (7)$$

where  $\mu$  is the reduced mass of the electron-hole pair and  $L$  is the confined dimension.

Mechanically, nanorods exhibit high stiffness and strength due to their small size and low defect density. The stress-strain relationship is given by:

$$\sigma = E \cdot \varepsilon \quad (8)$$

where  $\sigma$  is stress,  $E$  is Young's modulus, and  $\varepsilon$  is strain.

Nanorods also show polarization-dependent optical behavior and tunable electrical and thermal conductivity, making them suitable for sensors, LEDs, photodetectors, and drug delivery systems.

### 3.4 Behavior of HMM Substrate

Hyperbolic Metamaterials (HMMs) are artificially engineered media composed of subwavelength layers of metals and dielectrics, or metal nanowire arrays embedded in a dielectric host. These structures exhibit highly anisotropic permittivity, resulting in hyperbolic

dispersion relations not found in natural materials. The permittivity tensor of a uniaxial HMM is given by:

$$\varepsilon = \begin{pmatrix} \varepsilon_{||} & 0 & 0 \\ 0 & \varepsilon_{||} & 0 \\ 0 & 0 & \varepsilon_{\perp} \end{pmatrix} \quad (9)$$

where  $\varepsilon_{||}$  and  $\varepsilon_{\perp}$  are the in-plane and out-of-plane components, respectively. If  $\varepsilon_{||} \cdot \varepsilon_{\perp} < 0$ , the material supports hyperbolic dispersion.

From an optical manipulation perspective, HMMs are powerful because they support propagation of high-k modes (large wavevectors), which are typically evanescent in conventional media. This enables strong light-matter interaction, enhanced optical gradient forces, and deep subwavelength confinement.

The general dispersion relation for transverse magnetic (TM) waves in HMM is:

$$\frac{k_x^2 + k_y^2}{\varepsilon_{\perp}} + \frac{k_z^2}{\varepsilon_{||}} = \frac{\omega^2}{c^2} \quad (10)$$

This relation describes a hyperboloid when  $\varepsilon_{||} \cdot \varepsilon_{\perp} < 0$ , allowing HMMs to support an infinite density of photonic states. As a result, particles near HMM substrates experience enhanced optical gradient and scattering forces, making HMMs ideal platforms for optical trapping, pulling, and manipulation of nanoscale objects.

Moreover, the local field enhancement and anisotropic radiation channels provided by HMMs enable directional control over optical forces, even enabling pulling forces where particles move toward the source of light, contrary to traditional pushing behavior.

### 3.5 Force Analysis for Single and Clustered Nanorods

In the full-wave simulations carried out using the COMSOL Multiphysics numerical solver (where successful convergence of all the simulations, demonstrated in this article, has been attained), all optical force calculations were performed on the outer Gaussian surface enclosing the scatterers. By integrating the time-averaged Minkowski stress tensor over this closed surface, the total induced time-averaged radiation force exerted on the illuminated nanorods (by the generated fields) has been computed. Mathematically, this is shown below:

$$\vec{F}_{\text{Total}}^{\text{out}} = \oint [\bar{\mathbf{T}}^{\text{out}}] \cdot d\vec{s} \quad (11)$$

where  $\langle \bar{\mathbf{T}}^{\text{out}} \rangle$  is expressed as

$$\langle \bar{\mathbf{T}}^{\text{out}} \rangle = \frac{1}{2} \text{Re} \left[ \vec{\mathbf{D}}_{\text{out}} \vec{\mathbf{E}}_{\text{out}}^* + \vec{\mathbf{B}}_{\text{out}} \vec{\mathbf{H}}_{\text{out}}^* - \frac{1}{2} \bar{\mathbf{I}} \left( \vec{\mathbf{E}}_{\text{out}}^* \cdot \vec{\mathbf{D}}_{\text{out}} + \vec{\mathbf{H}}_{\text{out}}^* \cdot \vec{\mathbf{B}}_{\text{out}} \right) \right] \quad (12)$$

Here, “out” denotes the total exterior field of the scatterer (summation of incident field and the scattered field); E, D, B, and H refer to the electric fields, displacement vector, induction field vector, and magnetic field, respectively;  $\langle \rangle$  denotes the time average and  $\bar{\mathbf{I}}$  is the unity tensor.

### 3.6 Summary

Here explains the fundamental principles of optical forces, including gradient, scattering, and absorption forces, and details the rigorous calculation of these forces using Minkowski’s Stress Tensor. It also describes the unique optical and mechanical properties of nanorods and the anisotropic behavior of hyperbolic metamaterial (HMM) substrates that enable enhanced light-matter interactions. Finally, it outlines the methodology for force analysis on single and clustered nanorods via full-wave simulations.



## Chapter 4: Methodology

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This chapter outlines the comprehensive methodology adopted to investigate the generation of optical pulling forces on plasmonic, chiral, and dielectric nanorods positioned over a hyperbolic metamaterial (HMM) substrate. The study employs full-wave numerical simulations to evaluate the behavior of these nanorods under the influence of a time-harmonic plane-polarized laser beam. The methodology is structured around the configuration of nanorods, the design of the substrate, simulation tools, environmental conditions, and the mathematical computation of optical forces using Minkowski's stress tensor.

### 4.1 Simulation Environment and Tools

All simulations in this study are conducted using COMSOL Multiphysics 6.0, a finite element-based Multiphysics simulation software. The Radio Frequency (RF) module is used for solving the full-wave frequency domain Minkowski's equations. This environment enables detailed modeling of complex material structures such as multilayer HMMs and anisotropic nanostructures with nanometer-scale resolution.

COMSOL's essential modules utilized in this study include

- 3D electromagnetic wave simulation,
- Accurate meshing in high-index contrast regions,
- Custom material definition (chiral, plasmonic, dielectric),
- Boundary conditions to mimic open-space environments using Perfectly Matched Layers (PMLs).

### 4.2 Substrate Design and Material Selection

In this study, the substrate plays a critical role in mediating light-matter interactions and enhancing optical pulling forces. The chosen substrate is a planar multilayered hyperbolic metamaterial (HMM) constructed from alternating plasmonic (P) and dielectric (D) layers in a repeating P–D–P–D configuration, as shown in Figure 1. This periodic structure induces

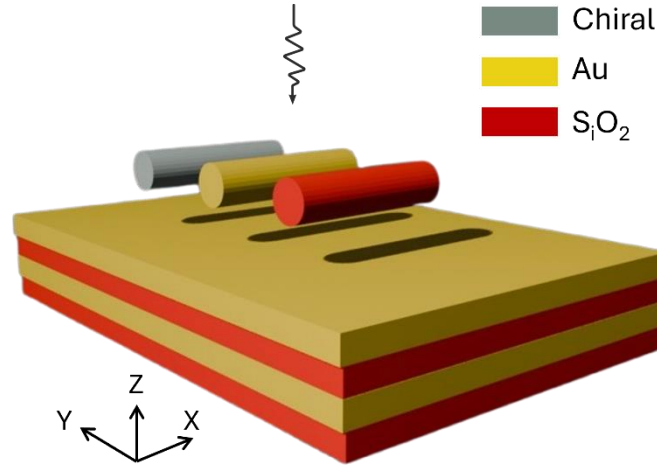


Figure 1: illustrates the proposed optical setup.

extreme optical anisotropy, resulting in a hyperbolic dispersion regime that supports a high density of photonic states (PDOS) and ultra-confined electromagnetic modes.

## Material Selection

### 1. Plasmonic Material (P):

- **Material:** Gold (Au)
- **Properties:** Strong negative real permittivity in the visible and near-infrared range, suitable for supporting surface plasmon polaritons (SPPs).
- **Optical Constants:** Taken from experimental data by Johnson and Christy [26].

### 2. Dielectric Material (D):

- **Material:** Silicon dioxide ( $\text{SiO}_2$ )
- **Properties:** Low-loss transparent dielectric with a relatively constant refractive index,  $n \approx 1.45$  in the optical regime.
- **Function:** Provides contrast in permittivity and enables anisotropy when alternated with gold.

### 3. Chiral Material:

- **Material:** Right-handed chiral dielectric
- **Properties:**
  - i. Relative permittivity,  $\epsilon_c = 1.45$
  - ii. Relative permeability,  $\mu_c = 1$
  - iii. Chirality parameter,  $k = +1$ , satisfying condition  $k^2 < \mu\epsilon$  for physical realizability.

➤ **Constitutive Relations:**

$$D = \epsilon_0 \epsilon_c E + ik\sqrt{\epsilon_0 \mu_0} H \quad (13)$$

$$B = \mu_0 \mu_c H - ik\sqrt{\epsilon_0 \mu_0} E \quad (14)$$

- **Function:** Enhances optical chirality and contributes to electric–magnetic dipole interference, supporting optical pulling especially for chiral and dielectric nanoparticles.

This composition ensures strong coupling of light into high-k modes and enhances near-field interactions necessary for producing optical pulling forces on the overlaid nanorods.

**Substrate Geometry and Parameters:**

- Each layer thickness:  $t = 20$  nm
- Each layer height and width = 200 nm
- Periodicity of the HMM:  $p = 40$  nm
- Number of Layers: 4
- Filling ratio:  $f = \frac{t}{p} = 0.5$

**Effective Medium Approximation**

To understand the optical response of the multilayer stack, the Effective Medium Theory (EMT) is employed. The anisotropic permittivity tensor components of the HMM are calculated as:

$$\epsilon_{||} = \frac{\epsilon_d \epsilon_m}{f\epsilon_d + (1-f)\epsilon_m} \quad (15)$$

$$\epsilon_{\perp} = f\epsilon_m + (1 - f)\epsilon_d \quad (16)$$

where  $\epsilon_m$ ,  $\epsilon_d$  permittivity of plasmonic and dielectric, respectively, and  $f$  is the filling ratio.

These expressions confirm the hyperbolic nature of the structure when  $\epsilon_{||} \cdot \epsilon_{\perp} < 0$ . In this regime, the substrate behaves as an open hyperboloid in k-space, allowing propagation of high spatial frequency modes that would otherwise be evanescent in natural materials.

## 4.3 Nanorod Configurations

### A. Cluster Configuration

In this setup, three types of nanorods, plasmonic (Au), chiral, and dielectric (SiO<sub>2</sub>) are arranged linearly over the HMM substrate. The plasmonic nanorod is placed at the center, with the dielectric and chiral rods positioned on the right and left, respectively. The distance between each nanorod is set at 25 nm to allow near-field coupling and dipole to dipole interactions.

### B. Single Nanorod Configuration

Each nanorod type is individually placed over the HMM substrate to observe intrinsic optical force behavior without the influence of neighboring scatterers.

#### Common Parameters:

- **Shape:** Cylindrical nanorod
- **Radius:**  $R = 30 \text{ nm}$
- **Length:**  $L = 150 \text{ nm}$
- **Elevation above HMM:** 30 nm

This configuration ensures anisotropy in scattering behavior and enhances interaction with polarized incident light.

## 4.4 Simulation Parameters

#### Light source parameters:

- **Type:** Time-harmonic plane-polarized laser beam.
- **Propagation Direction:** +Z to -Z axis
- **Polarization:** Magnetic field (H-field) polarized along the Y-axis.
- **Incident Angle:**  $\theta=0^\circ$

#### Wavelength Sweep:

- **Range:** 700 nm to 1600nm
- **Step Size:** 5 nm

#### Environmental Conditions:

- **Case 1:** Full in air medium
- **Case 2:** Half-air, half-water medium
- **Case 3:** Fully immersed in water (refractive index  $n = 1.33$ )

This range covers both sub-resonant and super-resonant regimes for the nanorods, enabling a comprehensive analysis of force directionality across wavelengths.

## 4.5 Calculation of Optical Forces

To quantitatively evaluate the optical forces acting on the plasmonic, chiral, and dielectric nanorods positioned above the hyperbolic metamaterial (HMM) substrate, employ the Minkowski Stress Tensor (MST) formalism. This method provides an accurate and complete representation of the time-average electromagnetic momentum flux interacting with the particle. The total time-averaged optical force  $F$  on each nanorod is calculated by integrating the MST over a closed Gaussian surface that encloses the nanorod:

$$F = \oint_S \mathbf{T} \cdot \hat{n} dS \quad (17)$$

where  $T$  is the stress tensor,  $\hat{n}$  is the outward-pointing normal vector to the surface  $S$ , and  $dS$  is the differential surface element. The MST tensor itself is defined as:

$$T_{ij} = \epsilon E_i E_j^* + \mu H_i H_j^* - \frac{1}{2} \delta_{ij} (\epsilon |E|^2 + \mu |H|^2) \quad (18)$$

Here,  $\epsilon$  and  $\mu$  denote the permittivity and permeability of the surrounding medium,  $E_i$  and  $H_i$  are the electric and magnetic field components, and  $\delta_{ij}$  is the Kronecker delta.

In the simulation environment (COMSOL Multiphysics), the integration is performed on a 3D Gaussian surface placed exactly 3 nm away from the nanorod boundary in all directions. This spacing ensures that the surface fully encloses the nanorod and captures the contributions from both incident and scattered electromagnetic fields while avoiding numerical singularities that may occur too close to the object. The chosen offset (3 nm) is a balance between proximity (to accurately reflect local near-field effects) and numerical stability in meshing and integration.

This MST-based approach enables us to resolve the magnitude and direction of the optical force along the Z-axis and determine whether the nanorod experiences a pushing (positive) or pulling (negative) optical force in different environmental and material conditions.

## 4.6 Summary

This chapter presents a detailed methodology for analyzing optical pulling forces on plasmonic, chiral, and dielectric nanorods above a hyperbolic metamaterial substrate using full-wave simulations in COMSOL. It describes the simulation setup, material and geometric

configurations, and environmental conditions. Optical forces are calculated through the Minkowski Stress Tensor by integrating the electromagnetic momentum flux over a closed surface surrounding each nanorod.

## Chapter 5: Results and Discussion

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This chapter explores the optical pulling effect exerted on nanorods of varying material types, chiral, plasmonic, and dielectric, under different optical configurations and environments. By leveraging a hyperbolic metamaterial (HMM) substrate and conducting full-wave simulations, the chapter investigates the emergence and sustainability of optical tractor beam (OTB) behavior. The study covers multiple scenarios, including changes in nanorod positioning, nanorod-to-substrate and inter-nanorod distances, size variations, and the absence of a substrate, to validate the robustness and generality of the proposed setup. Furthermore, the underlying physical mechanisms responsible for optical pulling in each nanorod type are thoroughly analyzed, highlighting the roles of field-induced dipole interactions, current density reversals, and gradient forces.

### 5.1 Result Analysis

A linearly polarized time-harmonic plane wave incident on the three distinct nanorods with diverse optical setups has been observed in this work. A recent work was conducted on distinct Rayleigh particles to obtain a generic optical pulling setup where all objects (chiral, plasmonic, and dielectric particles) get optical pulling forces [27]. On the other hand, a research article [28], [29] tells that these particles do not show optical pulling force, which means only dielectric particles get optical pulling, but plasmonic and chiral particles show pushing and lateral force, respectively.

#### 5.1.1 Force Results on Cluster and Single Nanorods

The observation started on a clustered configuration's numerical computation of the optical forces, and later, on single configurations. Near-field optical pulling forces obtained on distinct nanorods having chiral, plasmonic, and dielectric materials in three different environments, such as air, half-air & half-water, and water medium, considering cases 1,2, and 3, respectively, in clustered configurations, also in a single configuration. To obtain an OTB (optical tractor beam), the hyperbolic metamaterial (HMM) is used as a substrate in the proposed setup. As seen in Figure 2(a(i)), where the optical cluster is encircled by an air medium, this is achieved by utilizing the hyperbolic metamaterial in the proposed setup.

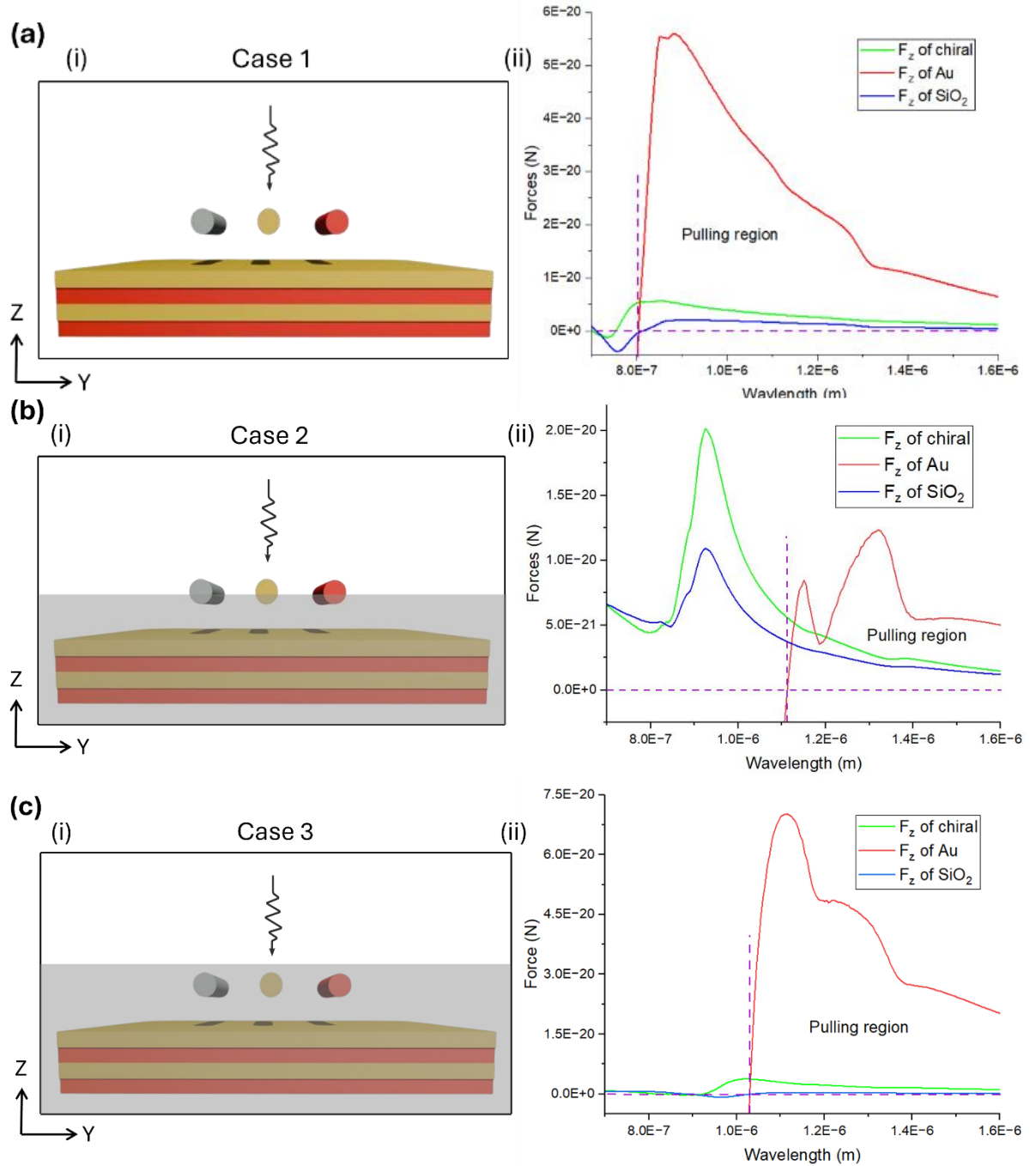


Figure 2: shows the schematic of the suggested optical setup in three distinct environments, together with the accompanying full wave simulation results. Case 1 (full air medium) is illustrated in (a), the setup is schematically shown in (i), and the full-wave simulation results for our suggested Case 1 configuration are shown in (ii). In fig.(b), A half-submerged water medium illustration of Case 2, (i) a schematic of the setup, and (ii) the full-wave simulation results for the configuration suggested for Case 2. In fig.(c), A schematic of the setup, (i) an illustration of Case 3 (completely submerged in water), and (ii) the outcomes of the full-wave simulation for our suggested Case 3 configurations.

The possibility of multiple scattering from the reflected beam of the NRs was removed by using



a homogeneous background medium and removing the substrate.

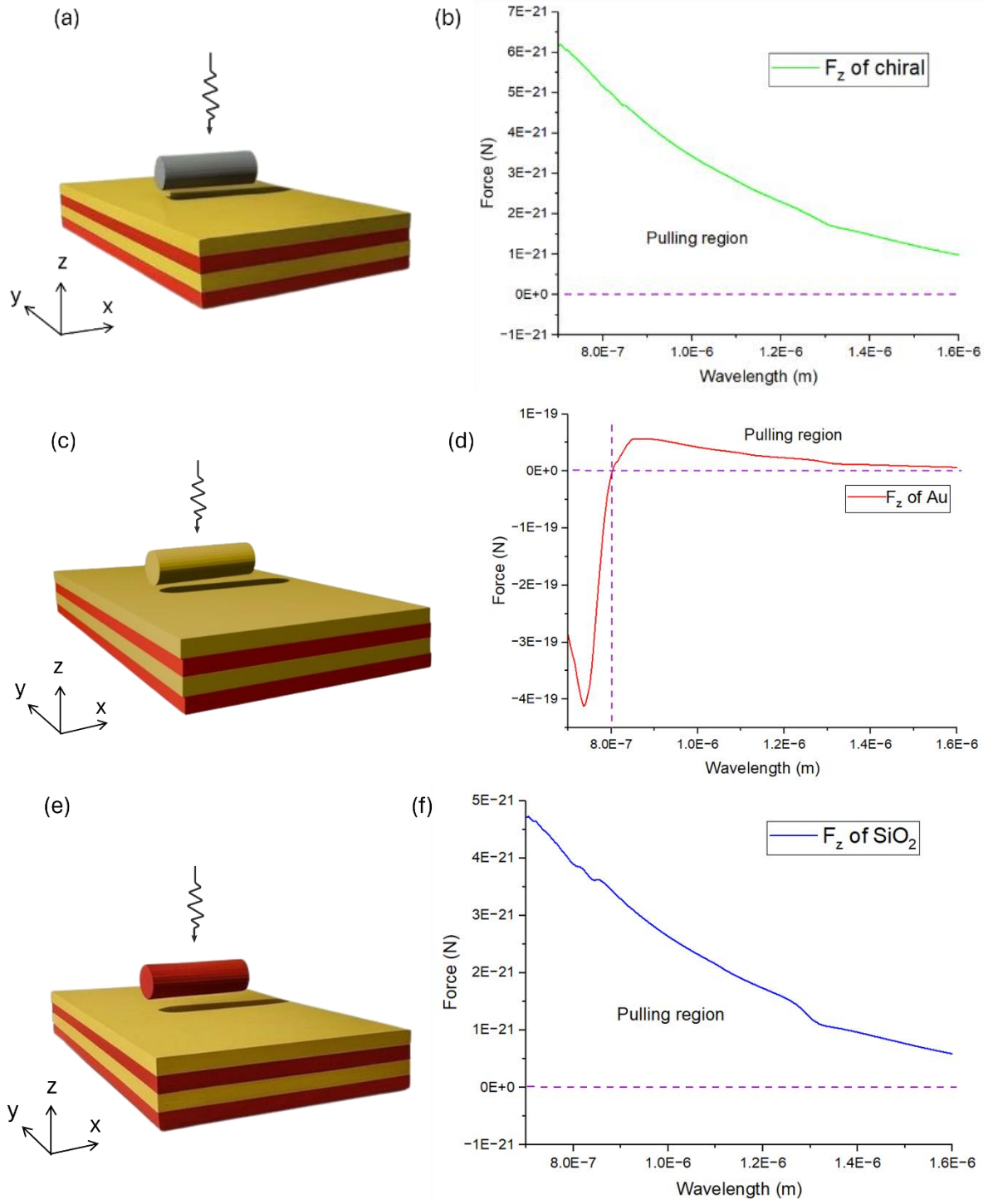


Figure 3: shows the schematic of proposed single optical configurations (a) for the chiral nanorod, (b) the full wave simulation results for the chiral nanorod, and (c) for the plasmonic nanorod, (d) the full-wave simulation results for the plasmonic nanorod (e) for the dielectric nanorod, and (f) the full wave simulation results for the dielectric nanorod.

However, as seen in Figure 2(a(ii)), the full-wave simulation clearly shows that all three NRs experience an optical pulling force starting at 800 nm, and this phenomenon continues for a wider wavelength range (up to 1600 nm). The optical pulling force is seen to be active from 1120 nm onwards in the half-air, half-water medium example (figure 2(b(i))), whereas the corresponding result of full wave simulation is shown in figure 2(b(ii)). Moreover, the OTB effect is active from 1030 nm onwards in the submerged water instance (Figure 2(c)), the dielectric nanorod gets an optical pushing force just a little bit in the non-pulling region. On the other hand, Chiral didn't get any other region of pulling force, which is explicitly shown in Figure 2(c(ii)), where the optical setup is shown in Figure 2(c(i)). It has been observed that this tugging effect has persisted up to 1600 nm in all three situations.

After observing the cluster configurations in diverse environments, the research work was further conducted on single configurations. As an additional proof of principle, It was also shown that, as illustrated in Figure 3, an optical pulling force is experienced by each nanorod of either chiral, plasmonic, or dielectric origin when it is positioned on the suggested HMM-based substrate. As seen in Figure 3(b & f), during the entire observation, optical pulling is enabled on both chiral and dielectric nanorods between 700 and 1600 nm. Additionally, the plasmonic particle has been observed to pull on the proposed system between 800 and 1600 nm, as shown in Figure 3(d). Furthermore, it should be mentioned that optical tugging was observed in the 800 to 1600 nm range when all three nanorods in a cluster were in the proposed optical setup. This indicates that the suggested data demonstrates similar pulling domains for both setups. It should be added that the optical setups for all these scenarios showed that optical pulling force was feasible when the setup for a single nanorod was put in different media, such as being completely submerged in water or half-submerged in it.

To ensure a more practical and generic optical setup for getting optical pulling forces of nanorods, this research work was conducted with all possible observations. A thorough discussion of the mechanisms that apply to all NRs, covering every scenario, is given in the subsection that follows:

### **5.1.2 Force Results on Interchanging Their Positions**

To examine how changing the location of a nanorod (NR) of one entity with another hinders the onset of the optical tractor beam effect, several parametric full-wave simulations have been carried out. Because it is impractical to maintain the precise arrangement of the NRs in a cluster, this is also crucial in substantiating the viability of the suggested configuration.

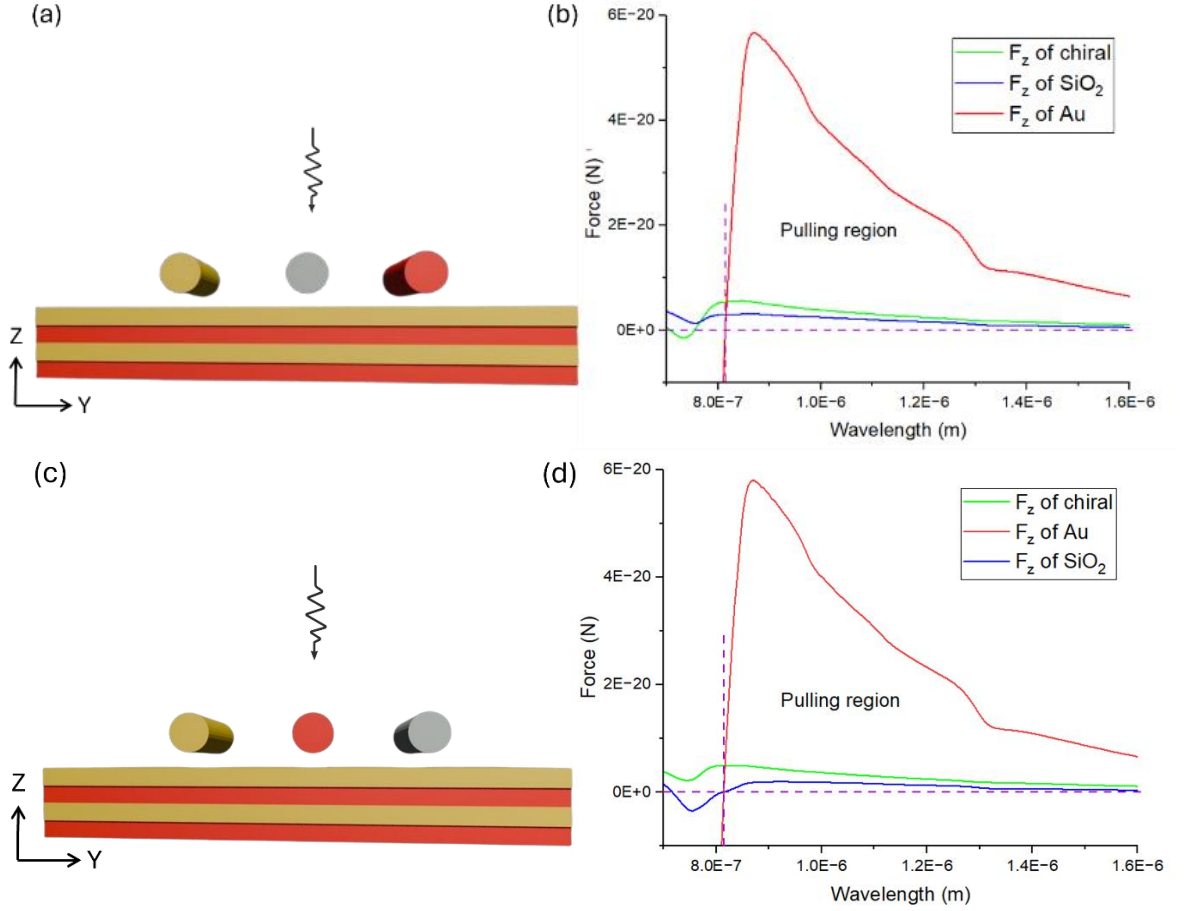


Figure 4: shows how NRs are positioned differently. The configuration with plasmonic, chiral, and dielectric (PCD) orientation is shown in two dimensions. (a). (b) The full-wave simulation's result for the PCD orientation. The proposed configuration with plasmonic, dielectric, and chiral (PDC) orientation is shown in two dimensions (c). (d) The PDC orientation's full wave simulation outcomes.

Therefore, as illustrated in Figure 4(a-d), full-wave simulations were performed after switching the nanorod locations. It should be mentioned that the heterogeneous cluster's optical pulling force is achieved regardless of where the NRs are positioned. Even if the NR's position has been altered, it is evident from looking at figure 4(b) & (d) that they continue to feel optical pulling force exactly from the same initial wavelength and consistently up to the end, 1600 nm. The only deviation caused by altering the NRs was the initial reverse behaviors of chiral and dielectric in the non-pulling region of the optical pulling region.

### 5.1.3 Force Results on Nanorods to Substrate Distances

Moreover, a parametric assessment of the optical force behavior by adjustment of the distance between the HMM and the NRs cluster. "h," which is the gap above the suggested HMM, has been adjusted to 30 nm in the proposed primary optical setup.

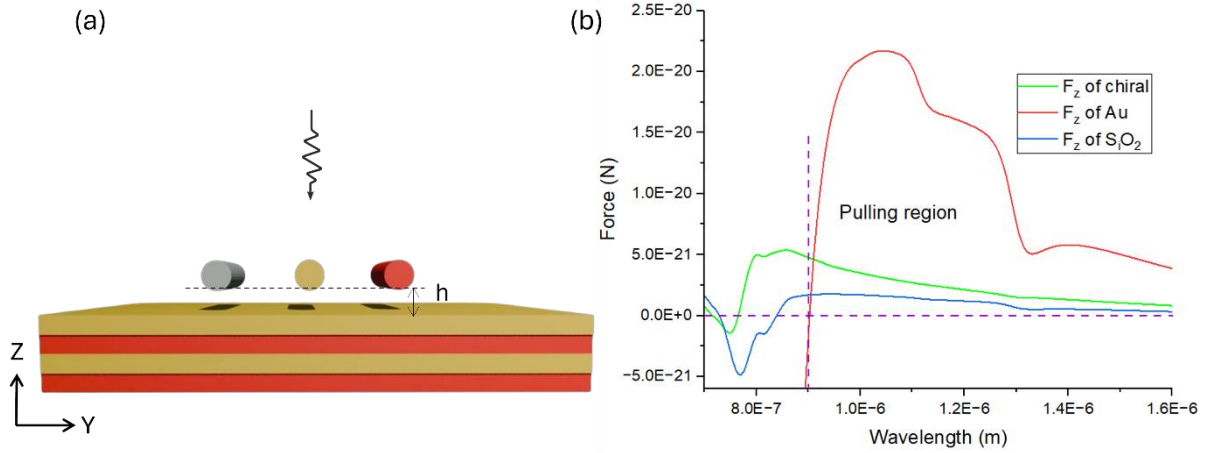


Figure 5: (a) shows the setup schematic for a gap of 20 nm around our suggested HMM, (b) and the full-wave simulation results.

That may not be the case, though, and as a result, it's seen that the optical pulling force when the gap above the suggested HMM is adjusted to  $h = 20$  nm, as seen in Figure 5.

### 5.1.4 Force Results on Nanorods to Nanorod Distances

The optical force behavior is evaluated parametrically by varying the distance between the cluster's NRs. A gap of 25 nm between the NRs above the proposed HMM has been arranged in the primary optical setup. That might not be the case, though, because when the distances between the nanorods are set between 50 and 100 nm, as shown in Figure 6, it might be observed that the optical pulling force. Interestingly, the optical pulling force of the heterogeneous cluster is obtained independently of the location of the NRs. Even though the NRs have been located far apart, it is clear from Figure 6(b) & (d) that they continue to experience optical pulling force precisely from the same beginning wavelength and continuously until 1600 nm. When both dielectric ( $\text{SiO}_2$ ) and chiral nanorods were positioned 50 nm and 100 nm from the plasmonic nanorod, as illustrated in Figure 6(a & c), they experienced optical pulling forces for the whole observation. But the dielectric experiences some pushing forces in the non-pulling zone; however, as Figure 6(b) illustrates, which is the

only slight variation of the  $\text{SiO}_2$  nanorod in the non-pulling zone, comparing both setups. the whole observation. But the dielectric experiences some pushing forces in the non-pulling zone; however, as figure 6(b) illustrates, which is the only slight variation of the  $\text{SiO}_2$  nanorod in the non-pulling zone, comparing both setups.

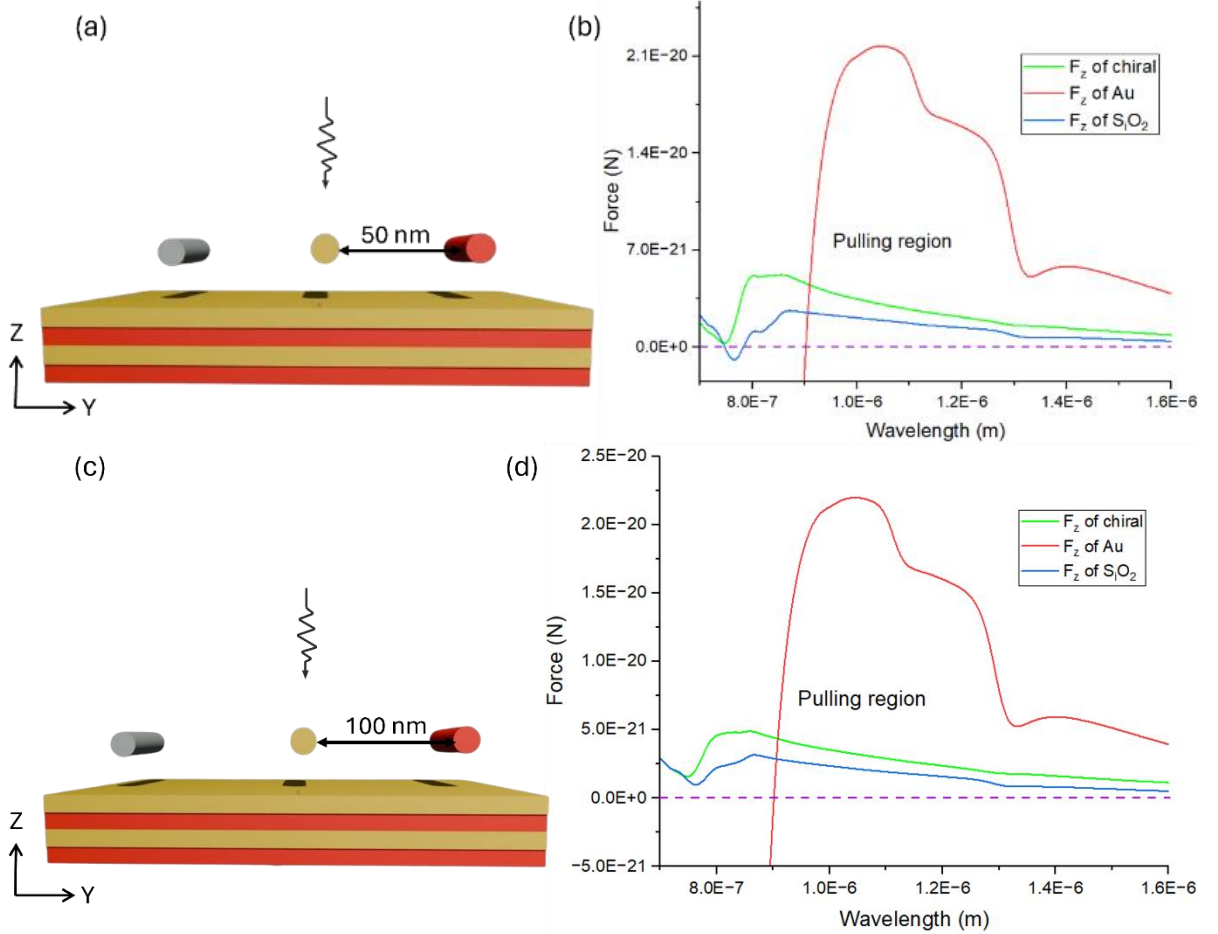


Figure 6: shows the corresponding results beside the optical setup's schematic. (a) For the gaps of 50 nm between the NRs over HMM, (b) depicts the respective full wave simulation results. (c) For the gaps of 100 nm between the NRs over HMM, (d) depicts the respective full wave simulation results.

### 5.1.5 Force Results on Radius Variations

Furthermore, a radius of 30 nm for the NRs in suggested optical setup; however, in a simulation setting, the NRs' radius can vary widely, so this setup simulated the setup for both a radius of 20 nm and a radius of 45 nm, as shown in figure 7 (a) and (c), respectively. The optical pulling for every NR in both scenarios is shown in Figure 7(b & d). It should be mentioned that changing the NRs' radii does not affect the heterogeneous cluster's ability to achieve optical pulling force. It is evident from looking at figures 7(b) and (d) that even if the NRs' radius has

been altered, they still feel an optical pulling force. The beginning wavelength for the optical pulling zone was the only deviation caused by altering the radius of NRs. Nevertheless, optical pulling has been accomplished up to 1600 nm for both examples, as seen in Figures 7(a) & (c).

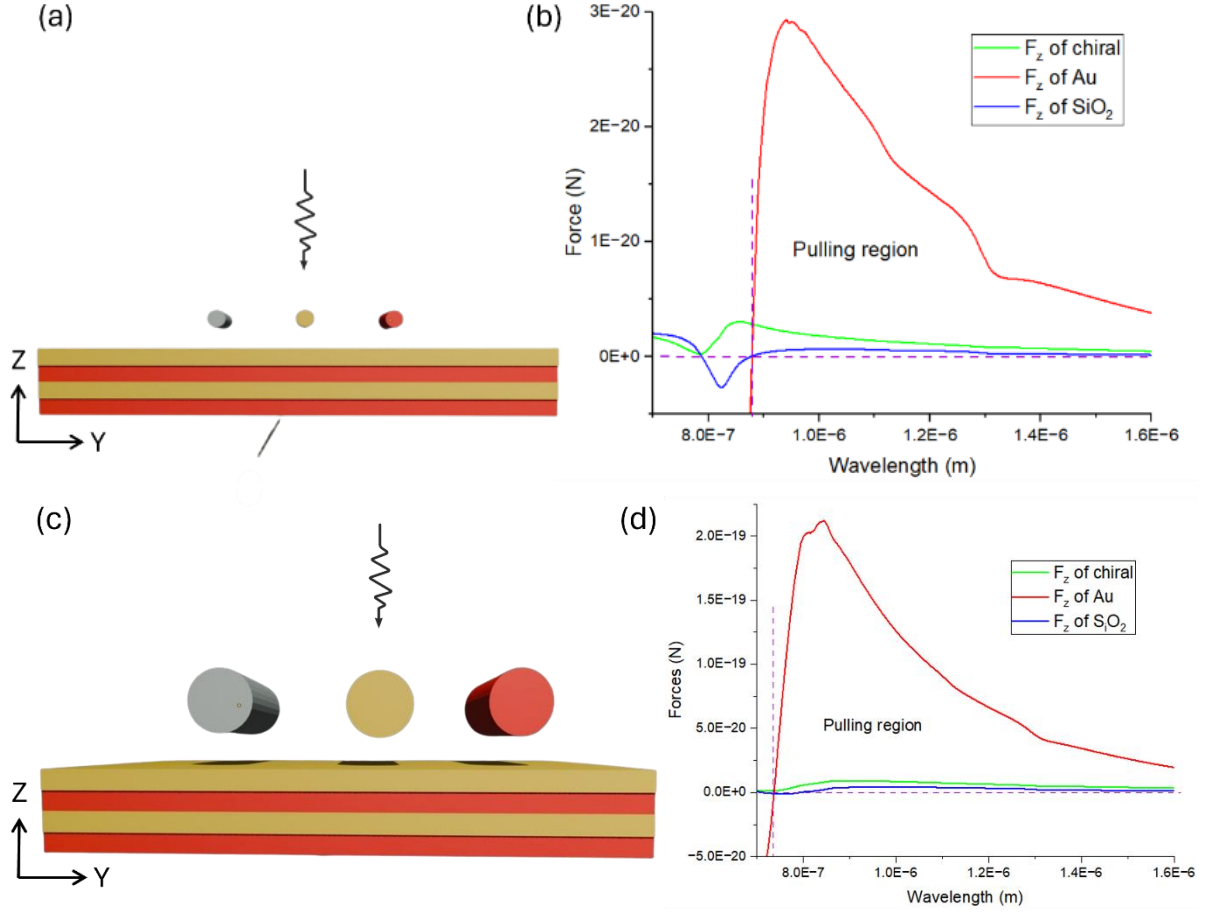


Figure 7: represents varying the radius of NRs. (a) The radius of NR is set to 20 nm; (b) depicts the respective full wave simulation results. While (c) the radius is set to 45 nm. (d) This depicts the respective full-wave simulation results.

### 5.1.6 Force Results on Length Variations

Additionally, a length of 150 nm for the NRs in the suggested optical setup; however, in a simulation setting, the NRs' lengths can vary widely, so the setup was simulated for both a length of 100 nm and a length of 250 nm, as shown in Figures 8 (a) and (c), respectively. We can achieve optical pulling for the NRs in both scenarios, which is shown in Figures 8(b) and (d). It should be mentioned that changing the NRs' length does not affect the heterogeneous cluster's ability to achieve optical pulling force for the plasmonic and chiral only, while it does affect the dielectric nanorod. It is evident from looking at Figure 8(b) and (d) that if the NRs' length has been altered, they feel an optical pulling force, except for the dielectric nanorod.

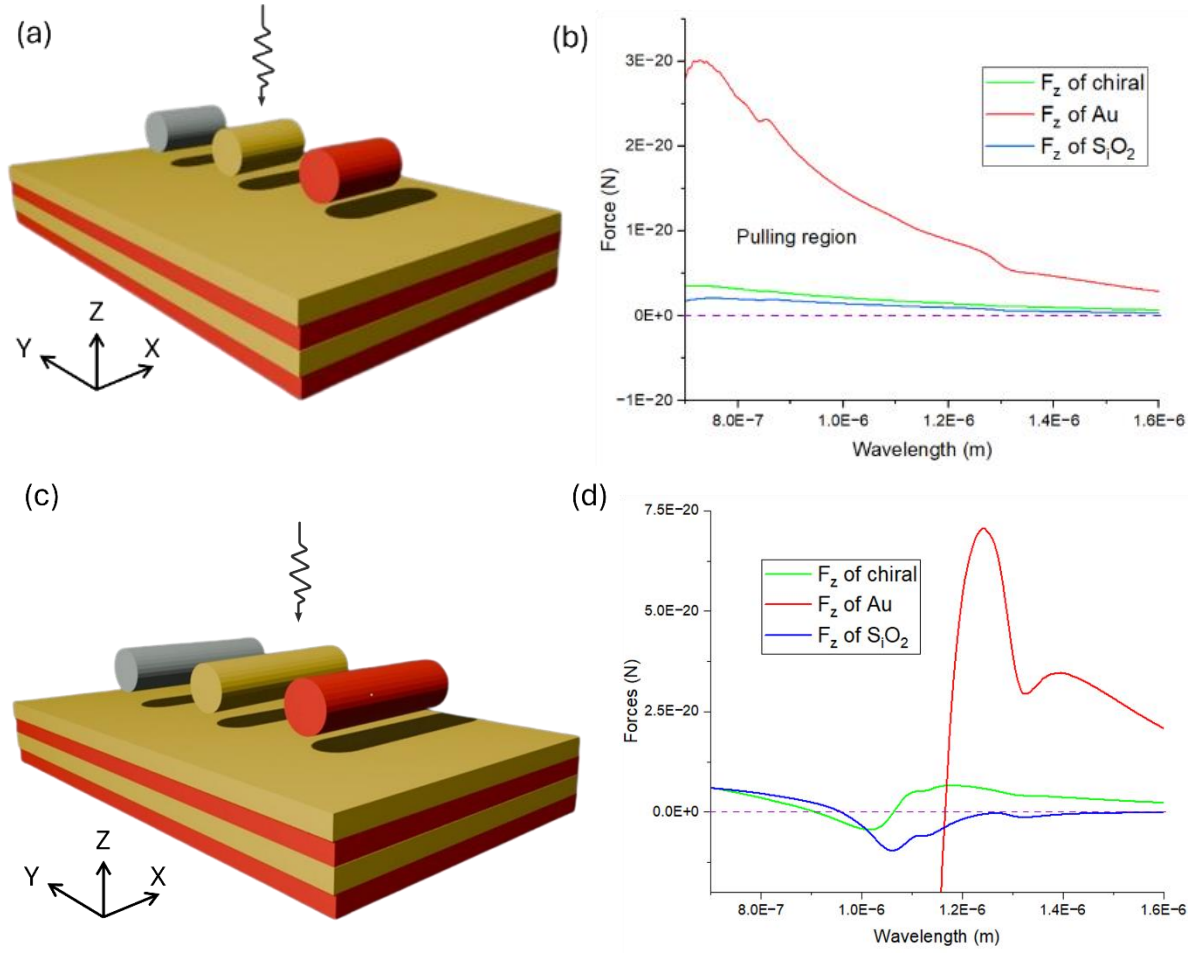


Figure 8: represents varying the length of NRs. (a) The length of NR is set to 100 nm; (b) depicts the respective full wave simulation results. While (c) the length is set to 250 nm. (d) the depicts the respective full wave simulation results.

The beginning wavelength for the optical pulling zone was the only deviation caused by altering the length of the NRs. Nevertheless, optical pulling has been accomplished up to 1600 nm for both examples, as seen in Figures 8(a) and (c).

### 5.1.7 Force Results on Size Variations

Lastly, Simulations were run to observe how the clustered nanorods (NRs) of different sizes behaved under optical force. The optical pulling forces were noticed when NRs had varying radii, as shown in Figure 9, even though the primary optical setup originally established 'R =



30 nm' as the radius of proposed NRs. The sizes of the chiral, plasmonic, and dielectric nanorods were fixed in Figure 9 (a, b) to 45 nm, 20 nm, and 30 nm, respectively.

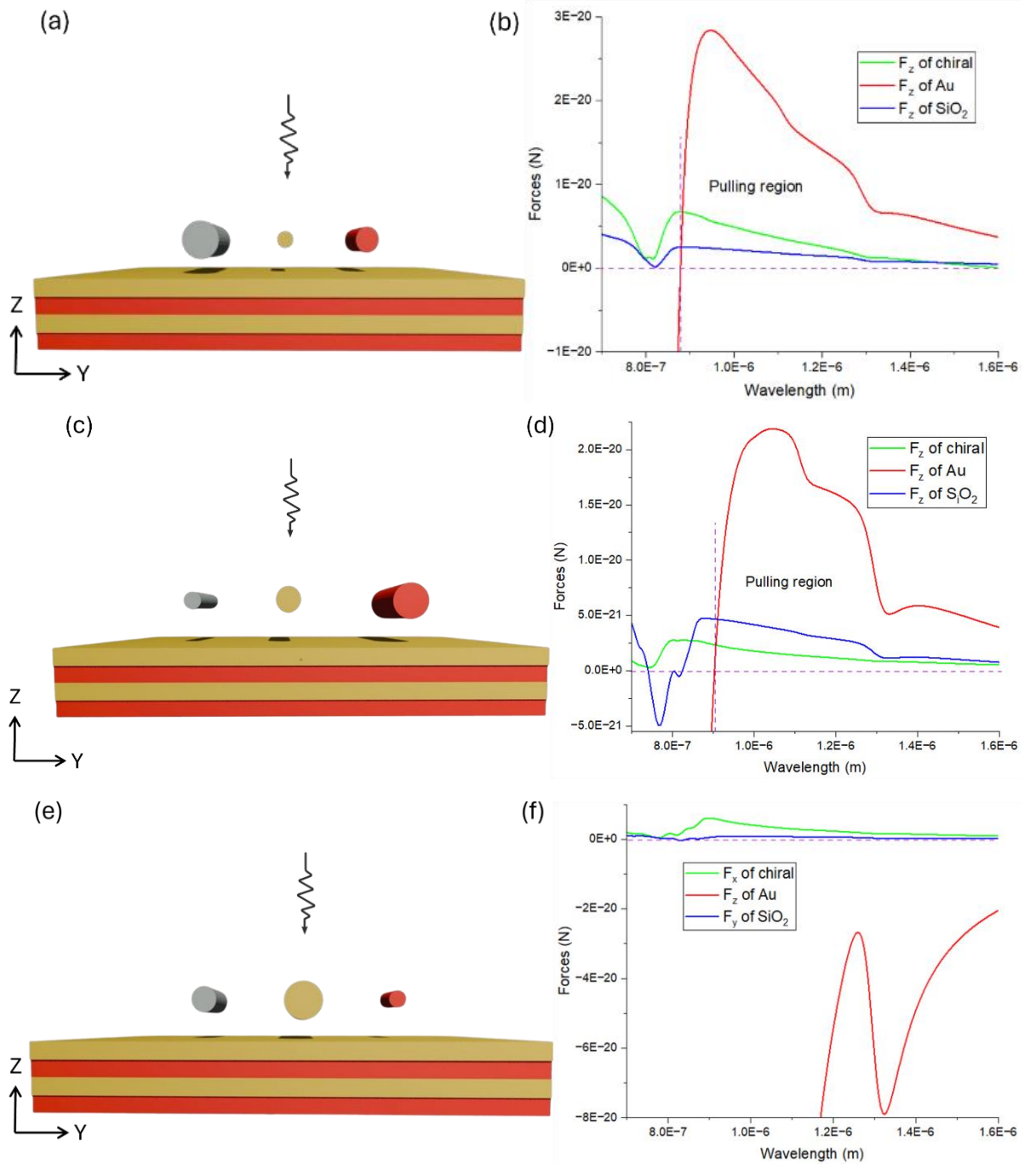


Figure 9: (a, c, e) The setup diagram and the full-wave simulation results for the cases where the NRs' radii were changed are shown in (b, d, f).

The optical pulling forces encountered by all three types of NRs ranged from 880 nm to 1600 nm. The chiral, plasmonic, and dielectric particle sizes were set to 20 nm, 30 nm, and 45 nm, respectively, in Figure 9 (c, d). In that case, optical pulling forces between 900 and 1600 nm were detected by all three kinds of NRs. The chiral, dielectric, and plasmonic nanorod



diameters for figures (e, f) were set at 30 nm, 45 nm, and 20 nm, respectively. Except for the plasmonic nanorod, two NRs in this case experienced optical pulling forces between 700 and 1600 nm.

### 5.1.8 Force Results on Without any Substrate

The optical force effect occurring on proposed clustered NRs without HMM present was simulated as of principal demonstration. Since initially eliminating the plasmonic-dielectric bilayer type HMM, as shown in Figure 10, which showed the setup and the full-wave simulation findings achieved, all the parameters for the suggested optical setup that were given in the section “Methodology” stayed the same.

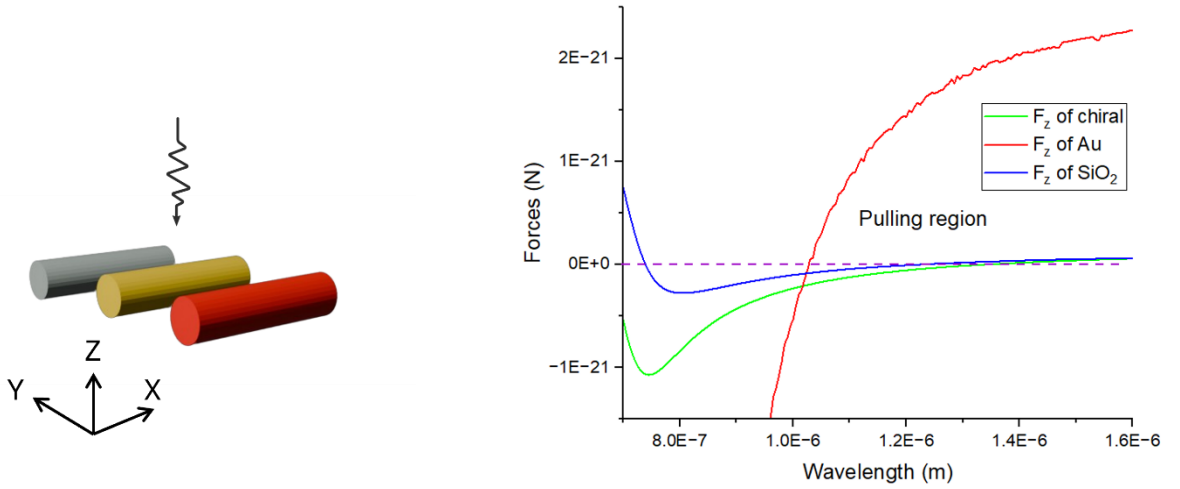


Figure 10: (a) displays the 3D schematic of the clustered NRs illuminated by a non-structured time-harmonic plane-polarized wave with a magnetic field polarization along the "Y" direction, propagating from "+Z" to "-Z," and an incident angle of  $0^\circ$  with the normal. (b) displays the findings of the full-wave simulation for the clustered NRs.

Only the plasmonic NRs exhibit the optical tractor beam effect, as can be plainly shown in Figure 10(b). In contrast to the force of tugging intended, NRs encountered an optical pushing force over the whole simulation domain.

## 5.2 Discussion

To see how various optical setup conditions might impact overall OTB behavior, multiple parametric simulations have also been carried out. Maintaining the precise optical configuration of the NRs in a cluster is impractical; hence, this scenario is essential to really demonstrate the viability of the suggested system.

### 5.2.1 Mechanism of optical pulling in a chiral nanorod

It is commonly known that chiral materials are drawn to light because of their chirality [30], [31]. In this case, the constitutive relation [32] defines the connection between the electric and magnetic momentum "p" and "m."

$$\mathbf{P} = \alpha_e \mathbf{E} + i_\chi \mathbf{H} \quad (19)$$

$$\mathbf{m} = -i_\chi \mathbf{E} + \alpha_m \mathbf{H} \quad (20)$$

where the complex electric polarizability, the magnetic polarizability, and the complex chiral polarizability are denoted by  $\alpha_e$ ,  $\alpha_m$ , and  $\chi$ , respectively. Investigating the contribution of the multipolar moments produced within the scatterer is crucial to understanding the optical pulling force applied to a chiral NR positioned above the suggested optical arrangement. This article discussed the electric and magnetic dipole moments because of their very small magnitude compared to the incident wavelength. More insight into how the chiral NR interacts with the incident light and experiences a pulling force may be gained by examining these instances. According to the article [31], [33] The total time-average optical force's mathematical expression can be represented, including the use of the polarizabilities, as follows:

$$\langle F \rangle = \langle F_e \rangle + \langle F_m \rangle + \langle F_{e-m} \rangle \quad (21)$$

$$\langle F_e \rangle = \frac{1}{4} \Re(\alpha_e) \nabla |\mathbf{E}|^2 + \frac{k}{2n} \Im(\alpha_e) \Re(\mathbf{E} \times \mathbf{B}^*) + \frac{1}{2} \Im(\alpha_e) \Im[(\mathbf{E}^* \cdot \nabla) \mathbf{E}] \quad (22)$$

$$\langle F_m \rangle = \frac{1}{4} \Re(\alpha_m) \nabla |\mathbf{B}|^2 + \frac{k}{2n} \Im(\alpha_m) \Re(\mathbf{E} \times \mathbf{B}^*) + \frac{1}{2} \Im(\alpha_m) \Im[(\mathbf{B}^* \cdot \nabla) \mathbf{B}] \quad (23)$$

$$\langle F_{e-m} \rangle = -\frac{k^4}{3} \sqrt{\frac{\mu}{\epsilon}} \Re(\alpha_e \alpha_m^*) \Re(\mathbf{E} \times \mathbf{B}^*) - \frac{k^3}{3} \mu \Im(\alpha_e \alpha_m^*) \times \frac{1}{2} \nabla |\mathbf{E}|^2 - \Re[(\mathbf{E}^* \cdot \nabla) \mathbf{E}] \quad (24)$$

$F_m$ ,  $F_e$ , and  $F_{e-m}$  are the magnetic dipole, electric dipole, and recoil force, respectively, in the above (22)-(24) equations, which can be written as,

$$F_e = \frac{1}{2} \Re(\nabla \mathbf{E}^*) \cdot \mathbf{p}$$

$$F_m = \frac{1}{2} \Re(\nabla \mathbf{B}^*) \cdot \mathbf{m}$$

$$F_{e-m} = -\frac{k^4}{12\pi\epsilon_0 c} \Re \mathbf{p} \times \mathbf{m}^*$$

The electric-magnetic dipole interference (third component in equation (21)) plays the main role in a chiral body gaining an optical pulling force, but the object can go into the forward direction (optical pushing force) with the help of momentum contribution from the electric and magnetic dipoles.

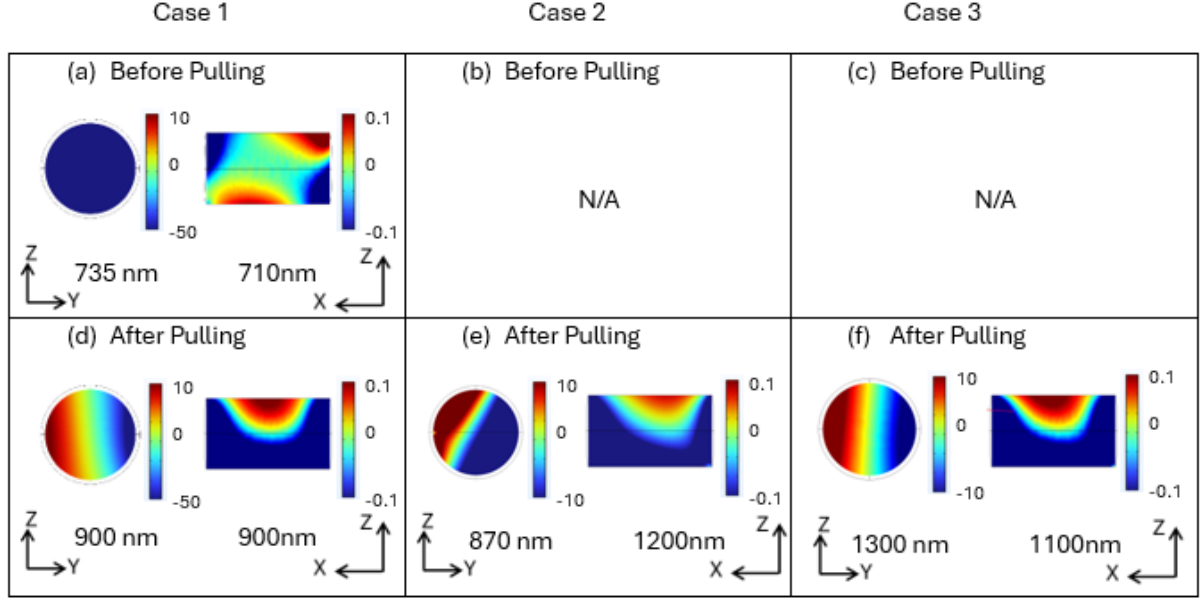


Figure 11: shows the electric and magnetic field profile of the chiral scatterer with various incident beam wavelengths in three different environments. (a & d) illustrates with the incident beams of 735 nm, 900 nm, and 710 nm, respectively, positioned on the proposed optical configuration in an air medium (case 1). However, the chiral scatterer, which is positioned on the same optical setup and partially submerged in water (case 2) at incident beam wavelengths of 870 nm and 1200 nm, respectively, is displayed in figure (e). The chiral scatterer positioned on suggested optical setup, completely submerged in water (case 3), at incident beam wavelengths of 1300 nm and 1000 nm, respectively, is finally shown in figure (f). As there is no other region except pulling, it goes to null in the figure (b-c). here, Y-Z and Z-X views show the E-field distribution (V/m), and H-field distribution (A/m), respectively.

By looking at Figure 11(a) and (d), which show the electric and magnetic field profiles of the chiral scatter positioned on proposed optical configuration in an air medium (case 1) with distinct wavelengths: 735 nm, 710 nm, and 900 nm, this can be seen that there is no discernible occurrence of an induced electric or magnetic dipole within the chiral NR cluster before the chiral nanorod (NR) experiences an optical pulling force. Nevertheless, the presence of electric and magnetic dipoles is visible in Figure 11(d) when the identical scatterer meets an optical pulling force. To provide an optical pulling force on a chiral object and contribute to the pertinent  $F(e-m)$  [31], [33], [34] fraction, equation (21) requires the presence of both electric

and magnetic dipolar counterparts. The interaction of the electric and magnetic dipole moments results in the recoil force, which also affects the intensity gradient force and scattering[34]. figure 11(e) and (f) clearly show the induced electric and magnetic dipoles within the pulling range. Therefore, it can be said that the main cause of the chiral NR's counterintuitive optical pulling force in the suggested optical setup is the object's simultaneous presence of electric and magnetic dipoles. Additionally, by looking at Figure 10, it can be said that the HMM also helps to enable the OTB impact on the chiral NR.

### **5.2.2 Mechanism of optical pulling in a plasmonic nanorod**

The analysis of the momentum contributions of the charges within the plasmonic nanorod is necessary to explain the origin of nonconservative optical pulling forces. One component of these originates from the forces acting on bound charges and currents, while the other is derived from the forces acting on free currents [35]. Free current force contributions can typically be disregarded in lossy media since they result in positive optical forces acting on the scatterers [36]. Naturally, the incidences of the SPP waves seen in Figure 1 apply forces along the -z direction, which is the direction of their propagation. To apply pushing forces on the plasmonic scatterers, the produced field always imparts momentum to the free currents. In active plasmonic devices, however, the distribution of induced current densities is connected to the force contribution from bound charges and currents. When comparing figures 12(a) and 12(b), the area where the force reversal occurs has seen a reversal of current density (within the gold nanorod). Similar conclusions can be drawn from comparing figures 12(a) and 12(c): current density reversal only happens when the force operating on the scatter shifts from pushing to a pulling type. This occurs because of the current density distribution remaining constant in the “before pulling” zone, flipping close to the force reversal, and then remaining reversed throughout the “pulling region” thereafter. It is well known that the force densities acting on bound charges and currents can be either positive or negative. The former gives rise to pushing pressures, while the latter creates optical tractor beams that draw plasmonic scatterers in the direction of the light source [37]. The Lorentz force distribution within a plasmonic object can likewise be used to explain this force reversal [38].

$$\langle F_{\text{total}} \rangle = \frac{1}{2} \text{Re} \left[ \oint \left( -\frac{1}{2} |E_{\text{Surf}}|^2 \nabla \varepsilon \right) ds \right] + \frac{1}{2} \text{Re} \left[ \int (\rho_{\text{free}} E_{\text{in}} + J_{\text{free}}(\text{in}) \times \mu_0 H_{\text{in}}) dv \right] \quad (25)$$

where, subscripts ‘surf’ and ‘in’ represent the surface and interior of the scatterer, respectively.

Now, substitute the value of  $F_{\text{total}}$  into equation (1), it can rewrite equation (9) as like below:

$$\oint \langle T^{\text{out}} \rangle \cdot ds = F_{\text{total}} = \langle F_{\text{Mink.}}^{\text{Surface}} \rangle + \langle F_{\text{Mink.}}^{\text{Bulk}} \rangle \quad (26)$$

where,

$$\langle F_{\text{Mink.}}^{\text{Bulk}} \rangle = \frac{1}{2} \text{Re} \left[ \int (\rho_{\text{free}} E_{\text{in}} + J_{\text{free}}(\text{in}) \times \mu_0 H_{\text{in}}) dv \right]$$

$$\langle F_{\text{Mink.}}^{\text{Surface}} \rangle = \frac{1}{2} \text{Re} \left[ \oint \left( -\frac{1}{2} |E_{\text{Surf}}|^2 \nabla \epsilon \right) ds \right] \quad (27)$$

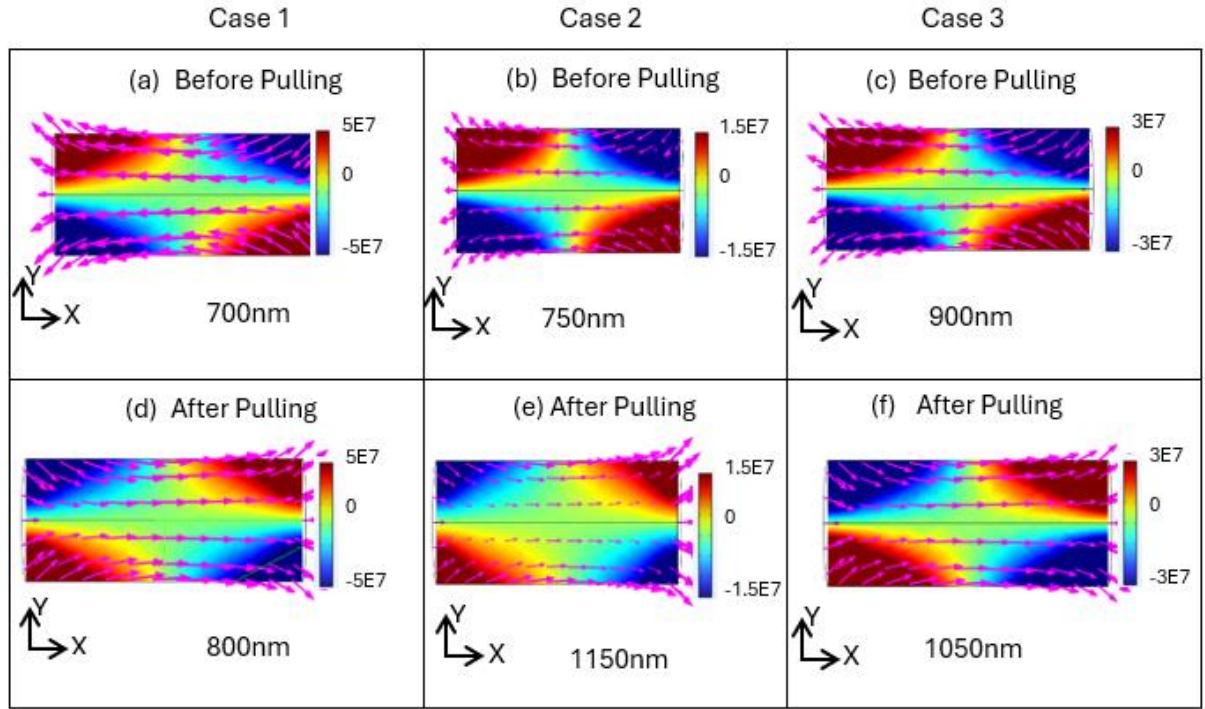


Figure 12: shows current densities (A/m<sup>2</sup>) (field and arrow profile) of plasmonic nanorod with various incident beam wavelengths in three different environments. The incident beams of 700 nm and 800 nm, respectively, placed on the suggested optical setup in an air medium (case 1) are used to demonstrate (a & d). On the other hand, figures (b & e) show the plasmonic scatterer located on the same optical setup and partially submerged in water (case 2) at incident beam wavelengths of 750 nm and 1150 nm, respectively. Finally, figures (c & f) display the plasmonic scatterer placed on the proposed optical setup, fully submerged in water (case 3), at incident beam wavelengths of 900 nm and 1050 nm, respectively. Here, the current density direction and field profile are displayed, allowing for a clear indication of reversal.

Because there are no free charge densities ( $\rho_{\text{free}}$ ) or free current densities ( $J_{\text{free}}(\text{in})$ ) within the particle, the bulk force of Minkowski, which is the second term in equation (26), exclusively applies to lossless dielectric objects at zero. The first term, the surface force of Minkowski,

also referred to as the Helmholtz force, is where the complete force for a lossless object originates [39]. One can make this force negative for a lossless item in several ways [39]. The bulk force is never zero for a lossy item, such as plasmonic particles, while the total force is dependent on both factors in equation (26). For lossy objects, the bulk force in equation (27) typically results in a positive bulk force since it is highly dependent on the free current density. In [36] ultimately, the total time-averaged pushing force for a plasmonic or absorbing particle is often determined by the prevailing bulk force component in equation (26). However, according to some recent findings [38-40] by adjusting both parts in equation (26), it is possible to attain the overall (total) pulling force on a lossy object. equation (27) provides a mathematical explanation of each term seen on the right side of equation (26). In scenario, when the plasmonic nanorod is subject to a pulling force, this force reversal is happening at the location where the plasmonic scatterer's current density is reversing (see equation (27): the bulk force of Minkowski). From figure 12(c) that the pulling zone, where the plasmonic scatterer encounters an optical pulling force, preserves the inverted profile.

This reversal is also readily apparent on the induced current density's arrow-surface plot, which is displayed in figure 12(a-f). It is well known that force densities on bound charges and currents can be either positive or negative; the former causes pushing forces to emerge, while the latter produces the optical tractor beam effect on the object, which pulls the plasmonic scatterer in the "+z" direction [41-42]. This reversal is the main factor contributing to the exertion of optical pulling force within the plasmonic object. It is noted that the orientation of the field profile remains the same (density field profile and arrow surface plot). Therefore, it is evident that the main factor for the application of an optical pulling force to a plasmonic scatter positioned above the suggested optical configuration is, in fact, the reversal of current density. The reversal is also seen in Figure 12(a-f), where the force acting on the plasmonic scatterer changes direction. This discovery has also been shown to be in complete agreement with the findings regarding the scenarios of a variable environment.

### **5.2.3 Mechanism of optical pulling in a dielectric nanorod**

To determine the source of optical tugging in a dielectric nanorod, an explanation must be provided using field-induced multipole moments. Equation (28) can be used to define the time average optical force produced by the incidence of a monochromatic optical field in a dielectric scatterer, typically accounting for the induced multipoles (up to the electric quadrupole) [43] as follows:

$$\langle F \rangle = F_p + F_m + F_{Qe} + F_{pm} + F_{Qep} \quad (28)$$

Contributions from electric dipoles, magnetic dipoles, electric quadrupoles, electric–magnetic dipole interference, and electric dipole–quadrupole interference are represented by the five terms in equation (28), respectively[44]. As seen below, the words above can be divided into two categories: one is caused by the beam's incidence, and the other is by direct interference [44-49]:

$$F = F_{\text{incident}} + F_{\text{interference}} \quad (29)$$

where,

$$F_{\text{incident}} = F_p + F_m + F_{Qe} \quad (30)$$

and,

$$F_{\text{interference}} = F_{pm} + F_{Qep} \quad (31)$$

The multipolar terms above can be expanded as like:

$$F_p = \frac{1}{2} \text{Re}[(\nabla E^*) \cdot p] \quad (32)$$

$$F_m = \frac{1}{2} \text{Re}[(\nabla B^*) \cdot m] \quad (33)$$

$$F_{Qe} = \frac{1}{4} \text{Re}[(\nabla \nabla E^*) \cdot Q_e] \quad (34)$$

$$F_{pm} = -\frac{k^4}{12\pi\epsilon_0 c} \text{Re}[p \times m^*] \quad (35)$$

$$F_{Qep} = -\frac{k^4}{40\pi\epsilon_0} \text{Re}[Q_e \cdot p^*] \quad (36)$$

The equations above identify the incident electric and magnetic field vectors as  $E$  and  $H$ , the induced electric dipole, magnetic dipole, and electric quadruple momentum as  $p = \alpha_e E$ ,  $m = \alpha_m H$ , and  $Q_e = \left(\frac{\gamma_e}{2}\right) (\nabla E + \nabla E^T)$ . Whereas  $\alpha_e = \frac{i6\pi\epsilon_0 a_1}{k^3}$ ,  $\alpha_m = \frac{i6\pi\epsilon_0 b_1}{\mu_0 k^3}$ ,  $\gamma_e = \frac{i40\pi\epsilon_0 a_2}{k^5}$  are the complex polarizabilities,  $a_1$ ,  $a_2$ ,  $b_1$  are the Mie coefficients, " $K$ " is the wave vector.

Now, we can rewrite the equation (28) as follows:

$$\begin{aligned} \langle F \rangle = & \frac{1}{2} \text{Re}[(\nabla E^*) \cdot p] + \frac{1}{2} \text{Re}[(\nabla B^*) \cdot m] + \frac{1}{4} \text{Re}[(\nabla \nabla E^*) \cdot Q_e] - \frac{k^4}{12\pi\epsilon_0 c} \text{Re}[p \times m^*] \\ & - \frac{k^4}{40\pi\epsilon_0} \text{Re}[Q_e \cdot p^*] \end{aligned} \quad (37)$$

or,

$$\langle F \rangle = \left( \frac{1}{2} \text{Re}[(\nabla E^*) \cdot \mathbf{p}] + \frac{1}{2} \text{Re}[(\nabla B^*) \cdot \mathbf{m}] + \frac{1}{4} \text{Re}[(\nabla \nabla E^*) \cdot \mathbf{Q}_e] \right) - \left( \frac{k^4}{12\pi\epsilon_0 c} \text{Re}[\mathbf{p} \times \mathbf{m}^*] + \frac{k^4}{40\pi\epsilon_0} \text{Re}[\mathbf{Q}_e \cdot \mathbf{p}^*] \right) \quad (38)$$

Using a single non-structured time-harmonic plane-polarized laser beam [i.e., in proposed case  $H = H_0 e^{jkz} \hat{y}$ ], a beating along the z-axis is formed. It creates a periodic change in intensity along the z-axis. As a result, a gradient force arises in the  $\mathbf{F}_{\text{incident}}$  contribution, which can be positive or negative. Now, this can express equation (38) as:

$$\langle F \rangle = \left( \frac{1}{2} \text{Re}[(\nabla E^*) \cdot \mathbf{p}] + \frac{1}{2} \text{Re}[(\nabla B^*) \cdot \mathbf{m}] + \frac{1}{4} \text{Re}[(\nabla \nabla E^*) \cdot \mathbf{Q}_e] \pm |F_g| \right) - \left( \frac{k^4}{12\pi\epsilon_0 c} \text{Re}[\mathbf{p} \times \mathbf{m}^*] + \frac{k^4}{40\pi\epsilon_0} \text{Re}[\mathbf{Q}_e \cdot \mathbf{p}^*] \right) \quad (39)$$

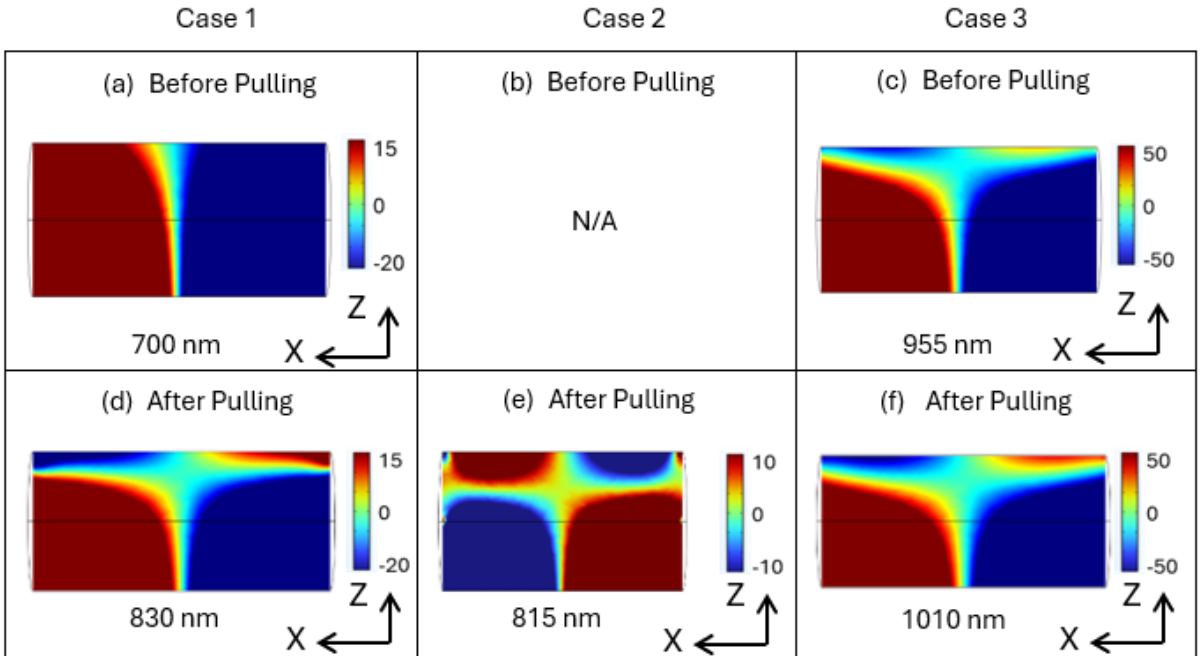


Figure 13 shows electric field distributions (V/m) of a dielectric nanorod with various incident beam wavelengths in three different environments. The incident beams of 700 nm and 830 nm, respectively, are placed in suggested optical setup in an air medium (case 1), as demonstrated in figure (a) and (d). On the other hand, figures (b & e) show the dielectric scatterer located on the same optical setup and partially submerged in water (case 2) at incident beam wavelengths of nm and 815 nm, where the figure (b) is null due to there being no other than pulling region over the full observation. Finally, figures (c & f) display the dielectric scatterer placed on proposed optical setup, fully submerged in water (case 3), at incident beam wavelengths of 955 nm and 1010 nm, respectively.



The gradient force is represented by the equation  $F_g$  above. According to equation (39),  $\epsilon - |F_g|$  is also in charge of producing the optical pulling force (in the "+z" direction) in addition to the term  $F_{\text{interference}}$ . An object's optical pulling force [50-51] is caused by  $\epsilon$  (negative gradient force). Optical pushing force (in the "-z" direction) is still produced by the other "usual" variables in  $F_{\text{incident}}$  [52]. Consequently, a periodic shift in intensity along the z-axis might cause the net force acting on a nanorod to be either positive or negative. The apparent pole generation within a dielectric scatter, situated close to the suggested optical setup, at wavelengths of 830 nm, 815 nm, and 1010 nm for cases 1, 2, and 3, respectively, is shown in Figure 13(d-f). From the electric field profiles, it can be observed that when the dielectric nanorod is subjected to an optical pulling force, electric dipole-quadrupole interference forms inside the object. According to equation (39), the momentum contribution from negative gradient force ( $-|F_g|$ ) and electric dipole quadrupole interference usually results in the application of optical pulling force in a dielectric object. Moreover, the development of electric dipoles when examining figure 13(a & c), which displays the electric field profiles for the identical object positioned over the suggested substrate at incident beam wavelengths of 700 nm (case 1) and 955 nm (case 3). From equation (39) that the electric dipole and positive gradient force ( $+|F_g|$ ) momentum contributions usually result in the application of the optical pushing force. Therefore, this deduces that the pulling force is mainly emerging in all circumstances because of the excitation of electric dipole-quadrupole interference and the negative gradient force ( $-|F_g|$ ). Consequently, the sum of the negative force and the negative interference component ( $F_{\text{interference}}$ ) is higher than the "normal" positive incident component,  $F_{\text{incident}}$ . The optical tractor beam effect is ultimately enabled when it causes the dielectric object to be dragged in the "+Z" direction. However, the electric dipole contribution and the local gradient force's dominance in the "-Z" direction, a positive gradient force  $F_g$  term emerges in the total force given in equation (39), cause an optical pushing force. The dielectric object is thus pushed in the "-Z" direction since the  $F_{\text{incident}}$  component grows larger than the  $F_{\text{interference}}$  portion.

### 5.3 Summary

This chapter confirms that chiral, plasmonic, and dielectric nanorods can achieve optical pulling forces using an HMM-based substrate across 700 -1600 nm. The effect remains consistent despite variations in rod position, size, and configuration. Each nanorod type

exhibits pulling due to distinct physical mechanisms, validating the robustness of the optical setup.

## Chapter 6: Conclusion and Future Works

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### 6.1 Conclusion

In this thesis, a simplified, practical, and universal optical setup utilizing a hyperbolic metamaterial (HMM) platform has been demonstrated to exert optical pulling forces on both single and clustered plasmonic, dielectric, and chiral nanorods. Through full-wave simulations, this confirmed that the proposed structure consistently induces pulling forces across three distinct environmental conditions: full air, half-immersed water, and fully immersed water. In the case of clustered nanorods, the optical pulling effect was observed within the wavelength range of 800–1600 nm, depending on the surrounding medium. For single nanorods, sustained pulling forces from 700 nm to 1600 nm, varying by nanorod material type. The underlying physical mechanisms for each nanorod type, dipolar interference, current density reversal, and multipole contributions were thoroughly analyzed. To the best of our knowledge, the ability to generate stable optical pulling forces across all three nanorod types and environmental media using a single hyperbolic platform is novel and may serve as a valuable foundation for future applications in nanomanipulation, photonic systems, and biosensing technologies.

### 6.2 Future Work

While this study successfully demonstrates a versatile optical pulling setup using hyperbolic materials (HMMs) across different media and nanorod types, several avenues remain open for further exploration:

**Experimental Validation:** Future research could focus on experimental realization of the proposed HMM structure and optical pulling setup. Fabrication of multilayered plasmonic dielectric substrates and integration with optical tweezing systems would provide practical insights and validate the simulation results.

**Thermal and Mechanical Stability:** Investigating the influence of thermal effects and mechanical stability of nanorods under high-intensity illumination would be essential, particularly for applications in biological systems or fluidic environments.

By addressing these aspects, future studies can move toward practical implementations of all-optical nanomanipulation platforms with high precision, adaptability, and application specific performance.

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